



CURRICULUM

Academic Year 2025– 2026

Batch 2023-2027

DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING

V & VI Semester B. E.

RAMAIAH INSTITUTE OF TECHNOLOGY
(Autonomous Institute, Affiliated to VTU)
BANGALORE – 54

About the Institute:

Dr. M. S. Ramaiah a philanthropist, founded 'Gokula Education Foundation' in 1962 with an objective of serving the society. M S Ramaiah Institute of Technology (MSRIT) was established under the aegis of this foundation in the same year, creating a landmark in technical education in India. MSRIT offers 18 UG programs and 15 PG programs. All these programs are approved by AICTE. All eligible UG and PG programs are accredited by National Board of Accreditation (NBA). The institute is accredited with '**A***' grade by **NAAC in March 2021** for 5 years. University Grants Commission (UGC) & Visvesvaraya Technological University (VTU) have conferred Autonomous Status to MSRIT for both UG and PG Programs since 2007. The institute has also been conferred autonomous status for Ph.D. program since 2021. The institute is a participant to the Technical Education Quality Improvement Program (TEQIP), an initiative of the Government of India. The institute has 380 competent faculty out of which 70% are doctorates. Some of the distinguished features of MSRIT are: State of the art laboratories, individual computing facility for all faculty members, all research departments active with sponsored funded projects and more than 300 scholars pursuing Ph.D. To promote research culture, the institute has established Centre of Excellence for Imaging Technologies, Centre for Advanced Materials Technology, Centre for Antennas and Radio Frequency systems (CARFS), Center for Cyber Physical Systems, Schneider Centre of Excellence & Centre for Bio and Energy Materials Innovation.

The Entrepreneurship Development Cell (EDC) and "Ramaiah Evolute" have been set up on campus to incubate startups. MSRIT has a strong Placement and Training department with a committed team, a good Mentoring/Proctorial system, a fully equipped Sports department, large air-conditioned library with good collection of book volumes and subscription to International and National Journals. The Digital Library subscribes to online e-journals from Elsevier Science Direct, IEEE, Taylor & Francis, Springer Link, etc. The Institute is a member of DELNET, CMTI and VTU E-Library Consortium. The Institute has a modern auditorium, recording studio, and several hi-tech conference halls with video conferencing facilities. The institute has excellent hostel facilities for boys and girls. MSRIT Alumni have distinguished themselves by occupying high positions in India and abroad and are in touch with the institute through an active Alumni Association.

As per the National Institutional Ranking Framework (NIRF), MoE, Government of India, M S Ramaiah Institute of Technology has achieved 75th rank among 1584 top Engineering Institutions & 31st Rank among 131 Schools of Architecture in India for the year 2025.

About the Department:

The Department of Information Science and Engineering (ISE) was established in the year 1992 with an objective of producing high quality professionals to meet the demands of the emerging field of Information Technology. Department offers Bachelor's program in Information Science and Engineering (B. E), Master's program in Data Sciences (MTech) and Doctoral program (Ph.D.). The Department of Information Science and Engineering, is a progressive department that has made significant contributions to Academics, Research and Innovation. Under Graduate (UG) is accredited by the National Board of Accreditation in 2001, 2004, 2010, 2015, 2018 and reaccredited in 2022 under Tier-1 till 2028. The department has highly qualified and competent faculty members committed to innovative teaching learning and quality research. Department has 8 well-equipped state of the art laboratories which meets the requirement of curriculum, innovation and research. Collaboration with industries such as Apple, Unisys, Mindtree, Intel, Google, SECO, IBM, NVIDIA etc., has a significant impact on the curriculum, computing infrastructure, teaching & learning and research. The curriculum is centered around Data Science, Artificial Intelligence, IOT, Cloud & Distributed Computing, System Programming, Computer Security and Software development. Curriculum and the teaching learning process ensure that the students demonstrate technical competence, ethical reasoning, creativity in identification & formulation of the problems and develop solutions by using appropriate tools & techniques. Department has established technical clubs/ professional student chapters to provide collaborative learning platform for the students. Echo system has been built to initiate start-ups/Innovation at the department level along with the mentorship program. The activities of the Department led to high profile placements, motivation to become an entrepreneur, and encouragement for higher learning.

VISION OF THE INSTITUTE

To be an Institution of International Eminence, renowned for imparting quality technical education, cutting edge research and innovation to meet global socio-economic needs

MISSION OF THE INSTITUTE

MSRIT shall meet the global socio-economic needs through

1. Imparting quality technical education by nurturing a conducive learning environment through continuous improvement and customization
2. Establishing research clusters in emerging areas in collaboration with globally reputed organizations
3. Establishing innovative skills development, techno-entrepreneurial activities and consultancy for socio-economic needs

QUALITY POLICY

We at MS Ramaiah Institute of Technology strive to deliver comprehensive, continually enhanced, global quality technical and management education through an established Quality Management System complemented by the synergistic interaction of the stake holders concerned

VISION OF THE DEPARTMENT

To evolve as an outstanding education and research center of Information Technology to create high quality Engineering Professionals for the betterment of Society

MISSION OF THE DEPARTMENT

Department of Information Science and Engineering shall create high quality IT Engineering Professionals for the betterment of society by:

1. Providing education through an ever improving curriculum and effective pedagogy techniques.
2. Encouraging extra and co-curricular activities to develop their overall personality along with technical skills.
3. Collaborating with industry and academia for strengthening research, innovation and entrepreneur ship ecosystem.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs):

PEO1: Become competent Information Technology professionals with continuous progress in career or learning.

PEO2: Productively engage with society by practicing research or entrepreneurship.

PEO3: Function effectively as professionals in a team environment or individually.

PROGRAM OUTCOMES (POs):

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs):

PSO1: Apply Mathematical models, programming paradigms and software development practices to solve real world problems

PSO2: Adopt computing and communication models for developing IT solutions.

PSO3: Acquire data engineering skills to develop intelligent systems in a multidisciplinary environment.

Semester wise Credit Breakdown for B.E Degree Curriculum Batch 2023-27

Semester	First	Second	Third	Fourth	Fifth	Sixth	Seventh	Eighth	Total Credits
Course Category									
Basic Sciences (BSC)	08	08	03	03	--	--	--	--	22
Engineering Sciences (ESC)	09	08	--	--	--	--	--	--	17
Humanities, Social Sciences and Management (HSMC)	02	02	--	--	03	03	--	--	10
Ability Enhancement Course (AEC)	01	02	01	01	01	--	--	--	06
Universal Human Values (UHV)	--	--	02	--	--	--	--	--	02
Professional Core Courses (PCC)	--	--	11	12	12	06	04	--	45
Integrated Professional Core Course (IPCC)	--	--	04	04	03		04	--	15
Professional Elective Courses (PEC)	--	--	--	--	03	06	03	--	12
Institutional Open Elective Courses (IOE)	--	--	--	--	--	03	03	--	06
Internship (INT)	--	--	--	--	--	--	--	05	05
Mini Project / Project Work (PW)	--	--	--	--	--	04	--	13	17
Non Credit Mandatory Courses (NCMC)	--	--	Yes	Yes	Yes	--	--	Yes	--
Skill Development Course (SDC)	--	--	--	--	--	01	02	--	03
Total Credits	20	20	21	20	22	23	16	18	160

SCHEME OF TEACHING V SEMESTER

Sl. No.	Subject Code	Subject	Teaching Department	Category	Credits				Total contact hours /week
					L	T	P	Total	
1	IS51	Software Engineering	ISE	PCC	3	0	0	3	3
2	IS52	Artificial Intelligence	ISE	IPCC	2	0	1	3	4
3	IS53	Computer Networks	ISE	PCC	4	0	0	4	4
4	IS54	Theory of Computation	ISE	PCC	2	1	0	3	4
5	ISE55x	Program Elective Course – 1	ISE	PEC	3	0	0	3	3
6	ISL56	Data Visualization Lab	ISE	PCC	0	0	1	1	2
7	ISL57	Computer Networks lab	ISE	PCC	0	0	1	1	2
8	AL58	Research Methodology & Intellectual property rights	ISE	HSMC	3	0	0	3	3
9	ISAEC59X	Ability Enhancement Course – V	ISE	AEC	1	0	0	1	1
Total					18	1	3	22	26
10	HS510	Environmental Studies *		NCMC	0	0	0	0	1

Programme Electives Course - 1

Sl.No	Subject Code	Subject
1	ISE551	Operations Research
2	ISE552	Mobile Application Development 1
3	ISE553	Computer Vision
4	ISE554	Distributed System
4	ISE555	Software design patterns

Ability enhancement Course - V

Sl.No	Subject Code	Subject
1	ISAEC591	NO SQL Database
2	ISAEC594	Frontend development using React JS

Nomenclature: **IPCC:** Integrated Professional Core Course, **PCC:** Professional Core Course, **HSMC:** Humanity and Social Science & Management Courses, **PEC:** Professional Elective Courses, **AEC**–Ability Enhancement Courses, **NCMC:** Non-credit Mandatory Course

L –Lecture, T – Tutorial, P- Practical/ Drawing

Note: ISE55x, where x=1,2,3,4,5

ISAEC59x, where x = 1,2,3...continued from previous

Integrated Professional Core Course (IPCC): Refers to Professional Theory Core Course Integrated with practical of the same course. Credit for IPCC is 03 and its Teaching–Learning hours (L : T : P) can be considered as (2 : 0 : 1). The theory part of the IPCC shall be evaluated both by CIE and SEE. The practical part shall be evaluated only by CIE (**no SEE**). However, questions from the practical part of IPCC can be included in the SEE question paper.

Professional Elective Courses: A professional elective (PEC) course is intended to enhance the depth and breadth of educational experience in Engineering and Technology curriculum. Multidisciplinary courses that are added to supplement the latest trend and advanced technology in the selected stream of engineering. Each group provides an option to select one course out of five courses. The minimum student's strength for offering professional electives is 10.

AICTE Activity Points to be earned by students admitted to BE program (For more details refer to Chapter 6, AICTE, Activity Point Program, Model Internship Guidelines):

Every regular student, who is admitted to the 4-year degree program, is required to earn 100 activity points in addition to the total credits earned for the program. Students entering 4 years degree program through lateral entry are required to earn 75 activity points in addition to the total credits earned for the program. The activity points earned by the student shall be reflected on the students VIII semester grade card. The activities to earn the points can be spread over the duration of the course. However, minimum prescribed duration should be fulfilled. Activity points (non-credit) have no effect on SGPA/CGPA and shall not be considered for vertical progression. Incase student fail to earn the prescribed activity points; VIII semester grade card shall be issued only after earning the required activity Points. Students shall be eligible for the award of degree only after the release of the VIII semester grade card.

SCHEME OF TEACHING VI SEMESTER

Sl. No.	Subject Code	Subject	Teaching Department	Category	Credits				Total contact hours /week
					L	T	P	Total	
1	AL61	Management & Entrepreneurship	ISE	HSMC	3	0	0	3	3
2	IS62	Machine Learning	ISE	PCC	3	1	0	4	5
3	ISE63x	Program Elective Course – 2	ISE	PEC	3	0	0	3	3
4	ISE64x	Program Elective Course – 3	ISE	PEC	3	0	0	3	3
5	ISL65	Machine Learning Lab	ISE	PCC	0	0	1	1	2
6	ISL66	DevOps Lab	ISE	PCC	0	0	1	1	2
7	ISOE0X	Open Elective - 1		IOE	3	0	0	3	3
8	ISP67	Mini Project	ISE	PW	0	0	4	4	-
9	IS68	Technical Skill Enhancement Course	Industry Expert	SDC	0	1	0	1	2
Total					15	1	6	23	23

Programme Electives Course – 2

Sl.No	Subject Code	Subject
1	ISE631	Mobile Application Development 2
2	ISE632	Internet of Things
3	ISE633	Block chain essentials and DApps
4	ISE634	System Simulation and Modeling

Programme Electives Course - 3

Sl.No	Subject Code	Subject
1	ISE641	Business Analytics
2	ISE642	Data Engineering and MLOps
3	ISE643	Natural Language Processing
4	ISE644	Cloud Computing

Institutional Open Elective - 1

Sl. No.	Subject Code	Subject
1	ISOE14	Data Analytics with Python
2	ISOE15	Fundamentals of Database Management Systems

Nomenclature, PCC: Professional Core Course, **PEC:** Professional Elective Courses, **IOE:** Institutional Open Elective, **PW:** Mini Project, **INT –** Internship

L –Lecture, T – Tutorial, P- Practical/ Drawing

Note: ISE63x , where x=1,2,3,4,5

ISE64x , where x=1,2,3,4,5

ISOE0x*, where x=1,2,.. continued from previous

L –Lecture, T – Tutorial, P- Practical/ Drawing/ Project work

Professional Elective Courses: A professional elective (PEC) course is intended to enhance the depth and breadth of educational experience in Engineering and Technology curriculum. Multidisciplinary courses that are added to supplement the latest trend and advanced technology in the selected stream of engineering. Each group provides an option to select one course out of five courses. The minimum student's strength for offering professional electives is 10.

Institutional Open Elective Courses:

Students belonging to a particular stream of Engineering and Technology are not entitled for the open electives offered by their parent department. However, they can take an elective offered by other departments, provided they satisfy the prerequisite condition, if any. Registration to open electives shall be documented under the guidance of the Proctor.

Selection of an open elective shall not be allowed if,

1. The candidate has studied the same course during the previous semesters of the program.
2. The syllabus content of open electives is similar to that of the Departmental core courses or professional electives.
3. A similar course, under any category, is prescribed in the higher semesters of the program.
4. The minimum students strength for offering open electives is 10.

Mini-project work: Mini Project is a laboratory-oriented course which will provide a platform to students to enhance their practical knowledge and skills by the development of small systems/applications. Based on the ability/abilities of the student/s and recommendations of the mentor, a single discipline or a multidisciplinary Mini- project can be assigned to a group having **not more than 4 students**. The CIE marks shall be awarded by a committee constituted by of the Head of the concerned department. The CIE marks awarded for the Mini-project work shall be based on the evaluation of project report, project presentation skill, and question and answer session as per the rubrics defined by the department. SEE will be conducted by the two examiners appointed by the Institute. SEE marks awarded for the mini project shall be based on the evaluation of project work report, project presentation skill and question and answer session.

Internship - All the students admitted shall have to undergo mandatory internship of 6 - 8 weeks during the intervening vacation of the IV & V semesters / intervening vacation of VI & VII semesters/ VII semester. **A Viva-Voce CIE examination** shall be conducted during VIII semester as per the rubrics defined by the department and the prescribed credits shall be included in VIII semester. Internship shall be considered as a head of passing and shall be considered for the award of degree. Those, who do not take-up/complete the internship shall be declared fail and shall have to complete during subsequent examination after satisfying the internship requirements.

AICTE Activity Points to be earned by students admitted to BE program (For more details refer to Chapter 6, AICTE, Activity Point Program, Model Internship Guidelines):

Every regular student, who is admitted to the 4-year degree program, is required to earn 100 activity points in addition to the total credits earned for the program. Students entering 4 years degree program through lateral entry are required to earn 75 activity points in addition to the total credits earned for the program. The activity points earned by the student shall be reflected on the students VIII semester grade card. The activities to earn the points can be spread over the duration of the course. However, minimum prescribed duration should be fulfilled. Activity points (non-credit) have no effect on SGPA/CGPA and shall not be considered for vertical progression. Incase student fail to earn the prescribed activity points; VIII semester grade card shall be issued only after earning the required activity Points. Students shall be eligible for the award of degree only after the release of the VIII semester grade card.

V SEMESTER

SOFTWARE ENGINEERING	
Course Code: IS51	Credits: 3:0:0
Pre – requisites: NIL	Contact Hours:42
Course Coordinator: Dr. Pushpalatha M N	

Course Contents

Unit 1

Introduction: Professional software development, Software engineering ethics, Case studies. Software processes: Software process models, Process activities, coping with change, The Rational Unified process. Agile Software Development: Agile methods, Plan-driven and agile development, Extreme programming. Agile Project Management and Scaling Agile Methods.

Recommended Resources / Links:

1. <https://www.youtube.com/watch?v=BBMEOagnBCU>
2. <https://www.youtube.com/watch?v=Z6f9ckEElsU>
3. <https://www.youtube.com/watch?v=vn6QvTb-TuQ>
4. <https://www.youtube.com/watch?app=desktop&v=KogKKAgN5Ps>

Unit 2

Requirements engineering: Functional and Non-functional requirements, the software requirements document, Requirements specification, Requirements Engineering Processes, Requirements elicitation and analysis, Requirements validation, Requirements management.

Recommended Resources / Links:

1. <https://www.youtube.com/watch?v=5cAxXUfb-KY>.

Unit 3

Architectural Design: Software Design and Implementation, Architectural design decisions, Architectural views, Architectural patterns, Application architectures. Design and implementation: Object-oriented design using the UML, Implementation issues, Open source development

Recommended Resources / Links:

1. <https://www.youtube.com/watch?v=8UlLgOf20Ho&list=PL4JxLacgYgqTgS8qQPC17fM-NWMTr5GW6>

Unit 4

A Perspective on Testing- Basic Definitions, Test Cases, Insights from a Venn Diagram, Identifying Test Cases- Specification-Based Testing, Code-Based Testing, Specification-Based versus Code-Based Debate, Fault Taxonomies, Levels of Testing, UNIT TESTING, Boundary Value Testing- Normal Boundary Value Testing, Robust Boundary Value Testing, Worst-Case Boundary Value Testing, Special Value Testing, Examples- Test Cases for the Triangle Problem, Test Cases for the NextDate Function, Equivalence Class Testing- Equivalence Classes, Traditional Equivalence Class Testing, Improved Equivalence Class Testing, Equivalence Class Test Cases for the Triangle Problem, Equivalence Class Test Cases for the NextDate Function.

Recommended Resources / Links:

1. <https://www.youtube.com/watch?v=TSacDWHGIWU>
2. <https://www.youtube.com/watch?v=egA7hCJnK-U>
3. <https://www.youtube.com/watch?v=zDMxb13ibqc>

Unit 5

Software Management: Project management: Risk management, Managing people, Teamwork. Project planning: Software pricing, Plan-driven development, Project scheduling, Agile planning, Quality management: Software quality, Software measurement and metrics.

Recommended Resources / Links:

1. <https://www.youtube.com/watch?v=b6mcMkQONic>
2. <https://www.youtube.com/watch?v=KqDlDubS-OU>

Text Books:

1. Ian Sommerville, Software Engineering, 9th Edition, Pearson Education, 2015.
2. Paul C. Jorgensen, Software Testing, A Craftsman's Approach, 4th Edition, Auerbach Publications, 2021

Reference Books:

1. Roger S. Pressman, Software Engineering-A Practitioners approach, 8th Edition, McGraw-Hill, 2014.
2. Shari Lawrence Pfleeger, Joanne M. Atlee, Software Engineering Theory and Practice, Third Edition, Pearson Education, 2006.
3. Waman S Jawadekar, Software Engineering Principles and Practice, Tata McGraw Hill, 2004.
4. Douglas Bell, Software Engineering for Students, A Programming Approach, 4th Edition, Pearson Education.

Course outcomes

At the end of the course, the student will be able to:

1. Understand the concepts of software engineering and development processes. (PO-1,8,9,10,11,12) (PSO-1)
2. Analyze the functional and non-functional requirements for the given problem (PO-1,2,9,10,11,12) (PSO-1)
3. Apply software architectural design for the given scenario (PO-1,2, 3,9,10,11,12)(PSO-1)
4. Understand Software testing and evolution processes. (PO-1, 9,10,11,12) (PSO-1)
5. Analyze Software Project Management issues and process improvement. (PO-1,2,11) (PSO-1)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2, CO3
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
Course Project	20	CO1, CO2, CO3, CO4, CO5
Semester-End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5

ARTIFICIAL INTELLIGENCE	
Course Code: IS52	Credits: 2:0:1
Pre – requisites: NIL	Contact Hours: 28L+14P
Course Coordinator: Dr. Anitha P	

Course Contents

Unit 1

Introduction: What is AI? **Intelligent Agents:** Agents and Environments, Rationality, The Nature of Environments, The Structure of Agents. **Problem-solving:** Problem-Solving Agents.

Recommended Resources / Links:

1. <https://nptel.ac.in/courses/112103280>
2. <https://nptel.ac.in/courses/106106126>
3. <https://media.dev.nptel.ac.in/content/mp4/112/103/112103280/MP4/mod01lec02.mp4>

Unit 2

Problem-solving: Example Problems, Search Algorithms, Uninformed Search Strategies, Informed Search Strategies: Greedy Best-First search, A* search, Heuristic functions:

Recommended Resources / Links:

1. <https://media.dev.nptel.ac.in/content/mp4/112/103/112103280/MP4/mod03lec08.mp4>
2. <https://media.dev.nptel.ac.in/content/mp4/112/103/112103280/MP4/mod03lec09.mp4>
3. <https://media.dev.nptel.ac.in/content/mp4/112/103/112103280/MP4/mod04lec11.mp4>

Unit 3

Adversarial Search: Optimal Decisions in Games, Heuristic Alpha-Beta Tree Search. **Logical Agents:** Knowledge Based Agents, The Wumpus World, Logic, Propositional Logic: A very simple logic.

Recommended Resources / Links:

1. <https://media.dev.nptel.ac.in/content/mp4/112/103/112103280/MP4/mod08lec23.mp4>
2. <https://nptel.ac.in/courses/106106140>
3. <https://media.dev.nptel.ac.in/content/mp4/112/103/112103280/MP4/mod04lec10.mp4>

Unit 4

First-Order Logic: Representation Revisited, Syntax and semantics of First-Order Logic, Using First - Order logic, Knowledge Engineering in First-Order Logic.

Inference in First Order Logic: Propositional Versus First Order Inference, Unification, Forward Chaining, Backward Chaining, Resolution.

Recommended Resources / Links:

1. <https://media.dev.nptel.ac.in/content/mp4/112/103/112103280/MP4/mod07lec18.mp4>
2. <https://nptel.ac.in/courses/106106143>
3. <https://media.dev.nptel.ac.in/content/mp4/112/103/112103280/MP4/mod12lec32.mp4>

Unit 5

Automated Planning: Definition of Classical Planning, Algorithms for Classical Planning.

Philosophy, Ethics and Safety of AI: The limits of AI, Can Machines Really Think? The Ethics of AI.

Recommended Resources / Links:

1. <https://media.dev.nptel.ac.in/content/mp4/107/106/107106090/MP4/mod01lec03.mp4>

2. <https://nptel.ac.in/courses/106105077>
3. <https://media.dev.nptel.ac.in/content/mp4/106/102/106102220/MP4/mod12lec92.mp4>

Laboratory Exercises

Implement and illustrate the working of the following AI problems in Python:

1. Implement Breadth-First Search (BFS) and Depth-First Search (DFS) to solve the 8-puzzle problem and compare their performance.
2. Implement Uniform Cost Search to find the least-cost solution for a path finding problem.
3. Implement the A* Search algorithm to solve the 8-puzzle problem using a suitable heuristic function.
4. Implement Minimax Algorithm for a simple game (e.g., Tic-Tac-Toe) and demonstrate optimal decision-making.
5. Enhance the Minimax Algorithm using Alpha-Beta Pruning and compare its efficiency with plain Minimax.
6. Implement a Knowledge-Based Agent for the Wumpus World using Propositional Logic.
7. Represent knowledge using First Order Logic for the Wumpus World and implement basic inference rules.
8. Implement and demonstrate the Hill Climbing Algorithm for solving a suitable search/optimization problem (e.g., N-Queens problem or mathematical function maximization). Analyze its performance and discuss the effect of local maxima, plateaus, and ridges."
9. Implement and demonstrate Travelling Salesman Problem (TSP) using heuristic approach.
10. Implementation of the problem-solving strategies: either using Forward Chaining or Backward Chaining.
11. Implement resolution principle on First Order Predicate Logic (FOPL) related problems.
12. Implement and demonstrate simple Chatbot with minimum 10 conversations.

Text Books:

1. Russell, S. J., & Norvig, P. (2021). Artificial Intelligence: A Modern Approach, Global Edition 4e.

Reference Books:

1. Elaine Rich, Kevin Knight, Shivashankar B Nair: Artificial Intelligence, 3rd Edition, Tata McGraw Hill, 2017.
2. Deepak Khemani "Artificial Intelligence", Tata Mc Graw Hill Education 2013.
3. Peter Jackson, "Introduction to Expert Systems", 3rd Edition, Pearson Education, 2007.

Course Outcomes:

1. Understand the fundamentals of Artificial Intelligence and apply problem-solving techniques using agents to solve example problems. (PO-1,2,3,5,12, PSO-1,3)
2. Analyze uninformed and informed search strategies in solving real world problems. (PO - 1,2,3,5,12, PSO-1,3)
3. Apply adversarial search methods to make optimal game decisions and construct knowledge-based agents using logical reasoning. (PO- 1, 2,3,5,12, PSO-1,3)
4. Apply knowledge using the syntax and semantics of First-Order Logic, and inference techniques such as unification, forward chaining, backward chaining, and resolution to solve reasoning problems. (PO- 1, 2,3,5,6,12, PSO-1,3)
5. Apply algorithms for classical planning in AI systems and analyze the philosophical, ethical, and safety implications of artificial intelligence. (PO- 1, 2,3,5,8, 12, PSO-1,3)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2, CO3
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
Lab experimental work & Record writing	10	CO1, CO2, CO3, CO4, CO5
NPTEL course/ Lab Test	10	CO1, CO2, CO3, CO4, CO5
Semester-End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5

COMPUTER NETWORKS	
Course Code: IS53	Credits: 4:0:0
Pre – requisites: -Nil-	Contact Hours: 56
Course Coordinator: Suresh Kumar K R	

Course Contents

Unit 1

Data communication Fundamentals: Introduction, Networks, Network types (LAN, WAN, MAN, Internet), Protocol Layering, TCP/IP protocol suite, OSI Model, **Physical layer:** Data and Signals, Basics of Periodic analog signals and digital signals, Transmission impairments, Data rate limits, Line coding schemes (unipolar, polar, bipolar) in digital-to-digital conversion.

Recommended Resources / Links:

1. <https://nptel.ac.in/courses/106105082>
2. <https://nptel.ac.in/courses/106105080>
3. <https://nptel.ac.in/courses/106105081>
4. <https://nptel.ac.in/courses/106105183>

Unit 2

Data Link layer: Introduction, Three types of addresses, ARP, Error detection and correction: Types of errors, redundancy, error detection vs error correction, block coding, cyclic codes - CRC and polynomials, checksum (Basic concept and internet checksum), **Data link control:** Framing, Media Access Control, Random Access Protocols, Controlled Access Protocols, Channelization.

Recommended Resources / Links:

1. <https://nptel.ac.in/courses/106105082>
2. <https://nptel.ac.in/courses/106105080>
3. <https://nptel.ac.in/courses/106106091>
4. <https://nptel.ac.in/courses/106105183>

Unit 3

Network layer: Network layer services, Introduction to connectionless service and connection oriented service, Congestion control, IPV4 Addresses (Address space, classful addressing, classless addressing, and NAT), Internet protocol (IP) datagram format, fragmentation, IP options, ICMP error reporting and query messages, Routing algorithms: Distance Vector Routing, Link State Routing, Multicast Distance Vector (DVMRP).

Recommended Resources / Links:

1. <https://a.impartus.com/ilc/#/course/81474/295>
2. <https://nptel.ac.in/courses/106105082>
3. <https://nptel.ac.in/courses/106106091>
4. <https://nptel.ac.in/courses/106105183>

Unit 4

Network Layer: IPV6 address representation and address space, The IPV6 Packet format, Extension header, Transition from IPV4 to IPV6, **Transport Layer:** Introduction, Transport layer Protocols (FSMs excluded)- Simple Protocol, Stop-and-Wait Protocol, Go-Back-N Protocol (GBN), Selective-Repeat Protocol, Bidirectional Protocols: Piggybacking.

Recommended Resources / Links:

1. <https://a.impartus.com/ilc/#/course/81474/295>
2. <https://nptel.ac.in/courses/106105081>
3. <https://nptel.ac.in/courses/106105183>

Unit 5

Transport Layer: Services, Port Numbers, User Datagram Protocol (UDP) – user datagram, services and applications, Transmission control Protocol (FSMs excluded): TCP Services and features, Segment, TCP Connection, Error control, and TCP Congestion control; Domain Name System (DNS) – Purpose and Resolution.

Recommended Resources / Links:

1. <https://a.impartus.com/ilc/#/course/83042/295>
2. <https://nptel.ac.in/courses/106106091>
3. <https://nptel.ac.in/courses/106105183>

Text Books:

1. Behrouz A. Forouzan – “Data Communications and Networking”, 5th Edition, Tata McGraw-Hill, 2017.

Reference Books:

1. Alberto Leon-Garcia and Indra Widjaja – “Communication Networks –Fundamental Concepts and Key architectures”, Second Edition, Tata McGraw-Hill.
2. William Stallings – “Data and Computer Communication”, Eight Edition, Pearson Education.

Course Outcomes:

1. Describe the fundamentals of data communication, network types, layered architectures (OSI and TCP/IP), and signal transmission techniques.
2. Understand the functioning of the Data Link Layer through addressing techniques, framing mechanisms, media access protocols and solve the errors induced during data transmission.
3. Use different algorithms to make the routing decisions, analyze IP addressing techniques and packet delivery mechanisms in IPv4 networks.
4. Differentiate between IPv4 - IPv6 protocols and explain transport layer communication through basic protocols.
5. Illustrate the working principles of UDP, TCP and the process of domain name resolution using DNS.

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components	Marks	Course outcomes attained
MCQ / Quiz / Surprise Test	10	CO1, CO2, CO3
Tool Demonstration	10	CO3, CO4, CO5
Semester-End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5

THEORY OF COMPUTATION	
Course Code: IS54	Credits: 2:1:0
Pre – requisites: Nil	Contact Hours: 28L + 28T
Course Coordinator: Dr. S R Manisekhar	

Course Contents

Unit 1

Finite Automata and Regular Expressions: Introduction to Finite Automata: central concepts of Automata theory, Deterministic finite automata, Nondeterministic finite automata, application of finite automata, Finite automata with Epsilon-transitions.

Recommended Resources / Links:

1. https://onlinecourses.nptel.ac.in/noc21_cs83/preview
2. <https://www.youtube.com/watch?v=9syvZr-9xwk>

Unit 2

Regular Languages, Properties of Regular Languages: Regular expressions; Finite Automata and Regular Expressions, Equivalence and minimization of automata.

Recommended Resources / Links:

1. https://onlinecourses.nptel.ac.in/noc21_cs83/preview

Unit 3

Context-Free Grammars and Languages: Context free grammars, Parse trees: Constructing parse trees, The yield of a parse tree, Applications, Ambiguity in grammars and Languages, Normal forms for CFGs.

Recommended Resources / Links:

1. https://onlinecourses.nptel.ac.in/noc21_cs83/preview
2. <https://www.youtube.com/watch?v=CuYZIsBSguw>

Unit 4

Pushdown Automata: The Pushdown automata: The languages of a PDA, Equivalence of PDA's and CFG's, Deterministic Pushdown Automata.

Recommended Resources / Links:

1. https://onlinecourses.nptel.ac.in/noc21_cs83/preview
2. [youtube.com/watch?v=Br44Zxv84-Q&pp=0gcJCfwAo7VqN5tD](https://www.youtube.com/watch?v=Br44Zxv84-Q&pp=0gcJCfwAo7VqN5tD)

Unit 5

Introduction to Turing Machine: The turning machine: Programming techniques for Turning Machines, Extensions to the basic Turning Machines, Turing Machine and Computers. Introduction to NP Hard and NP complete

Recommended Resources / Links:

1. https://onlinecourses.nptel.ac.in/noc21_cs83/preview
2. <https://www.youtube.com/watch?v=aQTdCUpNI6g>

TUTORIALS

- Practice the design of Deterministic Finite Automata problems.
- Design of Non- Deterministic Finite Automata problems.
- Construct ϵ -Non- Deterministic Finite Automata problems.
- Show the steps for converting NFA to DFA.
- Show the Converting step from ϵ -NFA to DFA problems.
- Construct Regular expression for the given a regular language.
- Design problems on Minimization of DFA.
- Construct Context free grammar given a context free language.
- Normal forms for CFGs
- Find the Ambiguity in grammars and Languages.

- Design Deterministic Push Down Automata problems
- Design Non Deterministic Push Down Automata problems
- Show the steps for converting PDA to CFG and Vice versa.
- Design of Turing Machine problems

Text Books:

1. John E. Hopcroft, Rajeev Motwani, Jeffrey Ullman: Introduction to Automata, Theory, Languages and Computation, 3rd Edition, Pearson education 2015

Reference Books:

1. John C Martin: Introduction to Languages and Automata Theory, 3rd Edition, Tata McGraw- Hill,
2. Daniel I.A. Cohen: Introduction to Computer Theory, 2nd Edition, John Wiley & Sons.

Course Outcomes:

At the end of the course, the student will be able to:

1. Design finite state machine using the concept of automata theory (PO-1,2,3,5,9,10,12; PSO-2)
2. Apply Regular Expression for the given finite Automata and vice versa. (PO-1,2,3,5,9,10,12; PSO-2)
3. Apply Context Free Grammar (CFG) for the given Normal Forms. (PO-1,2,3,5,9,10,12; PSO-2)
4. Design Push Down Automata for the given CFG and vice versa. (PO-1,2,3,5,9,10,12; PSO-2)
5. Design Turing Machine for a given language.(PO-1,2,3,5,9,10,12; PSO-2)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components	Marks	Course outcomes attained
MCQ / Surprise Test	10	CO1, CO2, CO3, CO4, CO5
Tool Demonstration	10	CO1, CO2, CO3, CO4, CO5
Semester-End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5

DATA VISUALIZATION LAB	
Course Code: ISL56	Credits: 0:0:1
Pre – requisites: Nil	Contact Hours: 14L
Course Coordinator: Dr. S R Mani Sekhar	

Course Contents

Programs on PowerBI or Tableau

1. Explore the functionality of PowerBI and Tableau
2. Exploring and Visualizing Sales Data. Import a sample sales dataset into Tableau. Create a dashboard that includes the following visualizations:
 - A bar chart showing total sales by product category.
 - A line chart showing monthly sales trends over the past year.
 - A geographic map showing sales distribution by region.
3. Advanced Visualization Techniques. Using a dataset on customer purchases, create a dashboard with the following advanced visualizations:
 - A scatter plot showing the relationship between customer age and total spending.
 - A heat map illustrating product sales intensity by store location.
 - A dual-axis chart comparing sales and profit margins over time.
4. Analyzing Sales Performance: Create various visualizations to analyze the sales performance.
 - Create a line chart to show sales trends over time.
 - Create a bar chart to compare sales across different regions.
 - Use a pie chart to display the distribution of sales among different product categories.
 - Create a dashboard combining these visualizations to provide an overall view of sales performance.
5. Financial Performance Dashboard: Create a comprehensive dashboard to monitor the financial performance of a company.
 - Create a combination of line and bar charts to compare actual revenue and expenses against budgeted figures.
 - Use a gauge chart to display key financial metrics such as net profit margin.
 - Create a waterfall chart to show the contribution of different factors to the overall profit.
 - Design an interactive dashboard that allows users to filter data by different time periods (e.g., monthly, quarterly, yearly).
6. Retrieve the employee performance prediction dataset from <https://docs.google.com/spreadsheets/d/19diCZNFaYPyA9nAXeVpipFYVRhhkl9Do/edit?usp=sharing&ouid=105051976182146717446&rtfpof=true&sd=true> in tableau .
 - Create a bar chart or treemap showing total sales revenue by product and region.
 - Create a scatter plot with discount rates on one axis and sales volume or revenue on the other. Add trend lines to identify correlations.
 - Use a dual-axis chart or side-by-side bar charts to compare sales and return rates across regions.
 - Use a heatmap or line chart to analyze monthly or quarterly sales and returns.

Programs on PySpark

7. Write a PySpark program to create a DataFrame with four columns: “name”, “age”, “city”, and

- “gender” and perform the following operations:
- Insert minimum 10 values for the given columns.
 - Filter rows with age greater than 30.
 - Add a new column named it “tax”.
 - Rename the “age” column to “years”.
 - Drop Multiple Columns from the given data frame.
8. Write a PySpark program to create a DataFrame containing information about various products, including ProductID, ProductName, Category, Price, StockQuantity, & Rating and perform the following operations:
- Insert minimum 10 values for the given columns.
 - Sort the DataFrame first by Price in descending order and then by Category in ascending order.
 - Find the total sales amount for each product by category.
 - Find the total sales amount and the total quantity sold for each product.
9. Using PySpark, analyze airline flight data (e.g., departure and arrival times, delays, carrier information) and perform the following operations:
- Load a CSV file containing airline flight data
 - Filter flights that were more than 15 minutes delayed.
 - Analyze whether there is any correlation between the flight length and the likelihood of a delay. Dataset: https://drive.google.com/drive/folders/1KTpKBf5w8VOyPNTTIniM9oN-pwaIgjqf?usp=drive_link
10. Consider airline flight data, given in the previous question. Perform the following operation using PySpark
- Group the data by airline carrier and compute the average delay for each one.
 - Determine the top five routes (origin-destination) with the highest average delay.
11. Given a Movie dataset containing user ratings for movies, using PySpark SQL perform the following operations
- Load a CSV file containing movie data
 - Create temporary views for movies and ratings.
 - Write queries to find the top 10 highest-rated movies with at least 10 ratings. Dataset: https://drive.google.com/file/d/17PFBafCd0J8brMNjdVv-NyNCBI01-7fr/view?usp=drive_link
 - Find the most active users (users who have rated the most movies).
12. Create a DataFrame containing real estate data with the following columns: HouseID, Location, Size, Bedrooms, Bathrooms, Price etc. Use the given dataset to build a linear regression model using PySpark's MLlib to predict the Price of a house based on the other features.
- Preprocess the data by handling missing values, encoding categorical variables (Location), and normalizing numerical features (Size, Bedrooms, Bathrooms, etc).
 - Split the data into training and testing sets.
 - Train a linear regression model on the training data.
 - Evaluate the model's performance on the test data using the root mean square error (RMSE)
 - Display the feature importances and interpret the results.
 - Dataset: <https://www.kaggle.com/datasets/sukhmandeepsinghbrar/house-prices-india/data>

Recommended Resources / Links:

1. <https://realpython.com/python3-object-oriented-programming/>
2. <https://python.swaroopch.com/oop.html>
3. https://python-textbok.readthedocs.io/en/1.0/Object_Oriented_Programming.html
4. <https://www.programiz.com/python-programming/>
5. <https://www.geeksforgeeks.org/python-programming-language/>

Text Books:

1. Chandraish Sinha, Mastering Power BI, 2nd edition 2024, BPB Publications
2. Arpana Chaturvedi, Prof Praveen Malik, Mastering Data Visualization with Tableau, 2024 BPB Publications
3. Andreas C Muller, Sarah Guido, Introduction to Machine Learning with Python, 2016 O'reilly Publications
4. Tomasz Drabas, Denny Lee, Learning PySpark, 2017 Packt Publishing

Reference Books:

1. Pramod Singh, Learn PySpark: Build Python-based Machine Learning and Deep Learning Models by Apress; 1st ed. edition

Course Outcomes:

At the end of the course, the student will be able to:

1. Comprehend various data visualization techniques and analyze methods for data pre- processing. (PO- 1, 2, 4)
2. Develop practical skills and expertise in utilizing data visualization tools. (PO – 3, 4, 5) (PSO – 1)
3. Apply data visualization techniques to solve real-world problems, making data-driven decisions in various domains such as business, healthcare, finance etc. (PO – 1, 2, 3, 4, 5) (PSO -1)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test	20	CO1, CO2, CO3
Other Components		
Conduction, Calculation, Record and Viva	30	CO1, CO2, CO3
Semester-End Examination (SEE)	50	CO1, CO2, CO3

COMPUTER NETWORKS LAB	
Course Code: ISL57	Credits: 0:0:1
Pre – requisites: -Nil-	Contact Hours: 14P
Course Coordinator: Suresh Kumar K R	

Course Contents

Part A: Simulation Using NS-2:

1. Simulate three nodes - point-to-point networks with duplex links between them. Set the queue size and vary the bandwidth and find the number of packets dropped
2. Simulate the different types of internet traffic such as FTP and TELNET over network and analyze the throughput
3. Simulate a four-node point-to-point network, and connect the links as follows: n0->n2, n1->n2 and n2->n3. Apply TCP agent between n0->n3 and UDP agent between n1->n3 agent. Changing the parameters, determine the number of packets sent by TCP and UDP.
4. Simulate the transmission of ping messages over a network topology consisting of 6 nodes and find the number of packets dropped due to congestion.
5. Simulate a bus topology with 5 nodes connected in sequence (n0–n1–n2–n3–n4). Configure a UDP agent with CBR traffic from node n0 to node n4. Analyze The packet flow along the bus, the delay and throughput observed at the receiver.
6. Simulate a bus topology with 4 nodes with duplex links. Generate FTP traffic over TCP between n0-n2 and CBR traffic over UDP between n1-n3. Analyze the throughput and packet delivery ratio.

Part B: Implement the following using C/C++:

7. Write a program for implementing the error detection technique for data transfer in unreliable network code using CRC (16-bits) Technique.
8. Given a graph of 5 nodes A to E, find shortest paths from any node as source to all other nodes using Distance vector routing.
9. Given a graph of 5 nodes A to E, find shortest paths from any node as source to all other nodes using Link state routing.
10. Packets from different flows arrive at a switch or router for processing. A good scheduling technique treats the different flows in a fair and appropriate manner. Implement priority queuing as a technique to improve Quality of Service.
11. Write a program to achieve Traffic management at Flow level by implementing Leaky Bucket Algorithm.
12. Using TCP/IP sockets, write a client-server program to make client send the file name and the server to send back the contents of the requested file name “sample.txt” with the following contents: “Hello we are at Computer Networks Lab” Display suitable error message in case the file is not present in the server.

Text Books:

1. Behrouz A. Forouzan – “Data Communications and Networking”, 5th Edition, 2017, Tata McGraw- Hill

Reference Books / Web links:

1. Alberto Leon-Garcia and Indra Widjaja – “Communication Networks –Fundamental Concepts and Key architectures”, Second Edition, Tata McGraw-Hill.
2. William Stallings – “Data and Computer Communication”, Eight Edition, Pearson Education.
3. NS2 manual - <https://networksimulator2.com/ns2-manual/>

Course Outcomes:

1. Simulate and analyze the network behavior against different parameters using NS2. (PO-1,2,3,4,5,9,11,12; PSO-2)
2. Design and implement the functionalities of different layers of the OSI model. (PO-1,2,3,4,9,12; PSO-2)
3. Interpret the results and produce the substantial document. (PO-9,10,12; PSO-2)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test	20	CO1, CO2, CO3
Other Components		
Conduction, Calculation, Record and Viva	10	CO1, CO2, CO3
Semester-End Examination (SEE)	50	CO1, CO2, CO3

OPERATIONS RESEARCH	
Course Code: ISE551	Credits: 3:0:0
Pre – requisites: Nil	Contact Hours: 42
Course Coordinator: Dr Savita K S	

Course Contents

Unit 1

Introduction to Operations Research (OR) and Linear Programming (LP): OR Models, Solving the OR Model, Phases of an OR Study; Modeling with Linear Programming (LP), Two variable LP Model, Graphical LP solution, Solution of a Maximization/Minimization Model, Computer solution with Excel Solver; LP Model in Equation Form, Transition from Graphical to Algebraic Solution; TORA Tool.

Recommended Resources / Links:

1. <https://www.youtube.com/watch?v=a2QgdDk4Xjw&list=PLjc8ejfjpgTf0LaDEHgLB3gCHZYcNtsoX&index=1>
2. <https://www.youtube.com/watch?v=pzbSURDWluA&list=PLjc8ejfjpgTf0LaDEHgLB3gCHZYcNtsoX&index=2>

Unit 2

The Simplex Method: The Simplex Method, Special cases in the Simplex Method, Degeneracy, Alternative optima, Unbounded solutions, Non-existing solutions; Artificial Variable Techniques: Two Phase Method, Big-M method; TORA Tool.

Recommended Resources / Links:

1. <https://www.youtube.com/watch?v=qxls3cYg8to&list=PLjc8ejfjpgTf0LaDEHgLB3gCHZYcNtsoX&index=4>

Unit 3

Duality and Game Theory: Introduction: Concept of Duality; Definition of Primal Dual Problems; General Rules for Converting any Primal into its Dual; Introduction to Dual Simplex Method; Computational Procedure of Dual Simplex Method; Illustrative Examples. Introduction of Game Theory, Characteristics of Games Theory; Minimax (Maximin) Criterion and Optimal Strategy; Saddle Point, Optimal Strategies and the value of game; Solution of Games with Saddle Point(s); Rectangular Games without Saddle Point; Arithmetic Method for (2x2) Games, Gambit tool for Game theory.

Recommended Resources / Links:

1. <https://www.youtube.com/watch?v=wrkcoRrsPQE&list=PLjc8ejfjpgTf0LaDEHgLB3gCHZYcNtsoX&index=10>

Unit 4

Project Management by PERT-CPM: Introduction, Applications of PERT/CPM Techniques, Basic Steps in PERT/CPM Techniques; Network Diagram representation, Rules for Drawing Network Diagram, Labeling Fulkerson's „I-J“ Rule; Time Estimates and Critical Path in Network Analysis; Project Evaluation and Review Technique; TORA tool

Recommended Resources / Links:

1. <https://www.youtube.com/watch?v=H58TPQNr2kM>

Unit 5

Transportation and Assignment Problems: Mathematical Formulation of transportation problem (TP); Matrix Form of TP; Feasible Solution, Basic Feasible Solution and Optimum Solution; Tabular

Representation; Special Structure of Transportation Table and their Problems; Initial Basic Feasible Solution to TP; Moving Towards Optimality; Degeneracy in TP; Unbalanced TP; Mathematical Formulation of Assignment Problem (AP); Fundamental Theorems; Hungarian Method for AP; TORA tool

1. Recommended Resources / Links:

1. <https://www.youtube.com/watch?v=Q31jKiEXxdc&list=PLjc8ejfjpgTf0LaDEHgLB3gCHZYcNtsoX&index=13>
2. <https://www.youtube.com/watch?v=RnZnIIksdwU&list=PLjc8ejfjpgTf0LaDEHgLB3gCHZYcNtsoX&index=14>
3. <https://www.youtube.com/watch?v=BUGIhEecipE&list=PLjc8ejfjpgTf0LaDEHgLB3gCHZYcNtsoX&index=16>

Text Books:

1. Operations Research: An Introduction – Hamdy A Taha – 11th Edition, Pearson Education, India, 2024.
2. Operations Research – S.D. Sharma –18th Edition, KNRN Publications, 2014.

Reference Books:

1. Introduction to Operations Research – Frederick S. Hillier and Gerald J. Lieberman – 10th Edition, Tata McGraw Hill, 2024

Course Outcomes:

At the end of the course students will be able to:

1. Formulate linear programming model for a given problem. (PO-1,2,3) (PSO-2,3)
2. Solve a linear programming model using appropriate methods. (PO-1,2,5,9,10,12) (PSO-2,3)
3. Apply game theory to model, analyze and solve the given problems. (PO-1,2,3,5,9,10,12) (PSO-2,3)
4. Analyze a Project network using PERT and CPM techniques. (PO-1,2,3,5,9,10,12) (PSO-2,3)
5. Apply transportation and assignment techniques to solve the given problems. (PO-1,2,3,5,9,10,12) (PSO-2,3)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2, CO3
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
Tutorial Evaluation	10	CO1, CO2, CO3, CO4, CO5
Test	10	CO1, CO2, CO3, CO4, CO5
Semester-End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5

MOBILE APPLICATION DEVELOPMENT - 1			
Course Code:	ISE552	Credits:	3:0:0
Pre – requisites:	Nil	Contact Hours:	42
Course Coordinator: Dr. S R Mani Sekhar			

Course Contents

Unit 1

Getting Started with App Development - basics of data, constant, variable, datatype, operators, and control flow in Swift, debugging, Xcode, building, running and debugging an app, Interface Builder, Swift strings, Guided project - Light.

Recommended Resources / Links:

1. <https://books.apple.com/us/book/develop-in-swift-fundamentals/id1511184145>
2. <https://www.apple.com/in/education/higher-education/app-development/>
3. Swift Programming Tutorial | FULL COURSE | Absolute Beginner -
<https://www.youtube.com/watch?v=CwA1VWP0Ldw>

Unit 2

Introduction to UI Kit – string, functions, structures, collections. UIKit—the system views and controls that make up a user interface and display data using Auto Layout and stack views.

Recommended Resources / Links:

1. <https://books.apple.com/us/book/develop-in-swift-fundamentals/id1511184145>
2. <https://www.apple.com/in/education/higher-education/app-development/>
3. Swift Programming Tutorial | FULL COURSE | Absolute Beginner -
<https://www.youtube.com/watch?v=CwA1VWP0Ldw>

Unit 3

More Swift –UIKit: Controls In Action, Loops, Structures, Classes, Inheritance, Closures, Variadics

Recommended Resources / Links:

1. <https://books.apple.com/us/book/develop-in-swift-fundamentals/id1511184145>
2. <https://www.apple.com/in/education/higher-education/app-development/>
3. Swift Programming Tutorial | FULL COURSE | Absolute Beginner -
<https://www.youtube.com/watch?v=CwA1VWP0Ldw>

Unit 4

Navigation and Workflows - build simple workflows and navigation hierarchies using navigation controllers, tab bar controllers, and segues, optionals and enumerations, type casting, Guided project- Personality Quiz

Recommended Resources / Links:

1. <https://books.apple.com/us/book/develop-in-swift-fundamentals/id1511184145>
2. <https://www.apple.com/in/education/higher-education/app-development/>
3. Swift Programming Tutorial | FULL COURSE | Absolute Beginner
<https://www.youtube.com/watch?v=CwA1VWP0Ldw>

Unit 5

Tables and Persistence – models, views, controllers, scroll views, table views, and building complex input screens, save data, work with images in the user's photo library. Guided project - List, a task- tracking app that allows the user to add, edit, and delete items in a familiar table- based interface.

Recommended Resources / Links:

1. <https://books.apple.com/us/book/develop-in-swift-fundamentals/id1511184145>
2. <https://www.apple.com/in/education/higher-education/app-development/>
3. Swift Programming Tutorial | FULL COURSE | Absolute Beginner -
<https://www.youtube.com/watch?v=CwA1VWP0Ldw>

Text Books:

1. Develop in Swift Fundamentals, Apple Books
2. Develop in Swift Data Collections, Apple Books

Course Outcomes:

At the end of the course, the student will be able to:

1. Design applications using swift as language and Xcode as IDE (PO – 1, 2, 3, 5) (PSO – 1, 2)
2. Create simple user interfaces using UIKit (PO – 1, 2, 3, 5) (PSO – 1, 2)
3. Design applications using advanced constructs of swift like class, structures, enums (PO – 1, 2, 3, 4, 5) (PSO – 1, 2)
4. Create multiple views and connect them using segues and navigation (PO – 1, 2, 3, 4, 5) (PSO – 1, 2)
5. Create apps using table view and scroll views (PO – 1, 2, 3, 4, 5) (PSO – 1, 2)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
MCQ / Surprise Test	10	CO1, CO2, CO3, CO4, CO5
Programming Assignment	10	CO1, CO2, CO3, CO4, CO5
Semester-End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5

COMPUTER VISION	
Course Code: ISE553	Credits: 3:0:0
Pre – requisites: nil	Contact Hours: 42
Course Coordinator: Dr. Shruthi G	

Course Contents

Unit 1

Introduction: Computer vision, Imaging modalities, Fundamental steps in image processing, Applications of computer vision. Digital Image Fundamentals: Image formation model, Sampling and quantization, Relationships between pixels. Mathematical tools used in image processing.

Recommended Resources / Links:

1. https://onlinecourses.nptel.ac.in/noc21_ee78/preview
2. https://onlinecourses.nptel.ac.in/noc23_ee39/preview

Unit 2

Spatial Filtering: Intensity transformation functions, Histogram processing (Histogram equalization, Histogram matching), Fundamentals of spatial filtering (Mechanics of spatial filtering, correlation and convolution), Smoothing spatial filters, Sharpening spatial filters.

Recommended Resources / Links:

1. https://onlinecourses.nptel.ac.in/noc23_ee39/preview
2. https://onlinecourses.nptel.ac.in/noc21_ee78/preview

Unit 3

Image Segmentation: Fundamentals, Detection of isolated points, line and basic edge, Thresholding, Region-based segmentation. Representation and Description: Representation (border following, chain codes, minimum-perimeter polygons) Boundary descriptors (simple descriptors, shape numbers), Region descriptors (simple descriptors, topological descriptors, texture).

Recommended Resources / Links:

1. https://onlinecourses.nptel.ac.in/noc23_ee39/preview
2. https://onlinecourses.nptel.ac.in/noc21_ee78/preview

Unit 4

Object Recognition: What Should Object Recognition Do? Feature, Geometric and semantic questions, Patterns and pattern classes, Recognition based on decision-theoretic methods, Matching, Optimum statistical classifier, Neural networks.

Recommended Resources / Links:

1. https://onlinecourses.nptel.ac.in/noc23_ee39/preview
2. https://onlinecourses.nptel.ac.in/noc21_ee78/preview

Unit 5

Morphological Processing: Erosion and Dilation, Opening and closing, Hit-or-miss transform, Morphological algorithms (Boundary extraction, Hole filling, Extraction of connected components). Compression Techniques: Fundamentals, Compression methods (Huffman, Arithmetic, Run-length coding)

Recommended Resources / Links:

1. https://onlinecourses.nptel.ac.in/noc23_ee39/preview
2. https://onlinecourses.nptel.ac.in/noc21_ee78/preview

Text Books:

1. Rafael C.Gonzalez, Richard E. Woods , “Digital Image Processing”, 4th ed., Pearson. 2018.
2. Computer Vision: A modern approach, D.A. Forsyth, J.Ponce, Pearson Education, 2016

Reference Books:

1. Anil K. Jain, “Fundamentals of Digital Image Processing”, Pearson Education, 2001.
2. B. Chanda and D. Dutta Majumdar , “Digital Image Processing and Analysis”, PHI, 2003.

Course Outcomes:

At the end of the course, the student will be able to:

1. Describe the fundamental concepts of a computer vision (PO-1) (PSO-3).
2. Apply spatial domain filters to improve the quality of the image (PO-1,2,3,5) (PSO-3).
3. Use segmentation, description, and recognition techniques for object identification in the image. (PO-1,2,3,5) (PSO-3).
4. Apply morphological operations and compression techniques in processing the image. (PO-1,2,3,5) (PSO-3).
5. Design solution for the identified problem by applying appropriate computer vision algorithm (PO-1,2,3,5,9,10,11,12) (PSO-3).

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2, CO3
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
Problem implementation	10	CO1, CO2, CO3, CO4, CO5
Report with Demo	10	CO1, CO2, CO3, CO4, CO5
Semester- End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5

DISTRIBUTED SYSTEM	
Course Code: ISE554	Credits: 3:0:0
Pre – requisites: NIL	Contact Hours: 42
Course Coordinator: Mr. Prashanth Kambli	

Course Contents

Unit 1

Characterization Of Distributed Systems: Introduction, Focus on resource sharing, Challenges

Remote Invocation: Introduction, Request-reply protocols, Remote procedure call, Introduction to Remote Method Invocation

Recommended Resources / Links:

1. nptel.ac.in/courses/106104182
2. https://www.youtube.com/watch?v=Azyizl9w2xo&list=PLrjkTql3jnm9FEOXHA_qjR-TMODlaIk-W

Unit 2

Distributed File Systems: Introduction, File service architecture,

Name Services: Introduction, Name services and the Domain Name System, Directory services

Recommended Resources / Links:

1. nptel.ac.in/courses/106104182
2. https://www.youtube.com/watch?v=Azyizl9w2xo&list=PLrjkTql3jnm9FEOXHA_qjR-TMODlaIk-W

Unit 3

Time And Global States: Introduction, Clocks, events and process states, Synchronizing

physical clocks, Logical time and logical clocks, Global states

Recommended Resources / Links:

1. nptel.ac.in/courses/106104182
2. https://www.youtube.com/watch?v=Azyizl9w2xo&list=PLrjkTql3jnm9FEOXHA_qjR-TMODlaIk-W

Unit 4

Coordination And Agreement: Introduction, Distributed mutual exclusion, Elections, Coordination and agreement in group communication, Consensus and related problems

Recommended Resources / Links:

1. nptel.ac.in/courses/106104182
2. https://www.youtube.com/watch?v=Azyizl9w2xo&list=PLrjkTql3jnm9FEOXHA_qjR-TMODlaIk-W

Unit 5

Distributed Transactions: Introduction, Flat and nested distributed transactions, Atomic commit protocols, Concurrency control in distributed transactions, Distributed deadlocks, Transaction recovery

Replication: Introduction

Recommended Resources / Links:

1. nptel.ac.in/courses/106104182

2. https://www.youtube.com/watch?v=Azyizl9w2xo&list=PLrjkTql3jnm9FEOXHA_qjR-TMODlaIk-W

Text Books:

1. George Coulouris, Jean Dollimore and Tim Kindberg, “Distributed Systems Concepts and Design”, Fifth Edition, Pearson Education, 2012.
2. Nancy A. Lynch, “Distributed Algorithms”, Hardcourt Asia Pvt. Ltd., Morgan Kaufmann, 2000.

Reference Books:

1. Andrew S. Tanenbaum, Maarten van Steen, “Distributed Systems Principles and Paradigms”, 2nd ed., Pearson Education, 2006.
2. Distributed Computing, Fundamentals, Simulations and Advanced topics, 2nd Edition, Hagit Attiya and Jennifer Welch, Wiley India

Course Outcomes:

At the end of the course, the student will be able to:

1. Understand the goals and challenges of distributed system (PO – 1) (PSO -2)
2. Demonstrate the remote invocation techniques for communication (PO – 1, 2, 3, 9, 10) (PSO -2)
3. Describe the architecture of distributed file systems and name services (PO – 1, 9, 10, 12) (PSO -2)
4. Apply clock synchronization algorithms to monitor and order the events. (PO – 1, 3, 9, 10) (PSO -2)
5. Analyze the performance of mutual exclusion, election and consensus algorithms. (PO – 1, 2, 9, 10, 12) (PSO -2)
6. Illustrate the fundamental concepts and algorithms related to distributed transactions and replication (PO – 1, 9, 10, 12) (PSO -2)

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2, CO3
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
Quiz	10	CO1, CO2, CO3, CO4, CO5
Assignment	10	CO1, CO2, CO3, CO4, CO5
Semester-End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5

SOFTWARE DESIGN PATTERNS	
Course Code: ISE555	Credits: 3:0:0
Pre - requisite: Object Oriented Programming, Software Engineering	Contact Hours: 42
Course Coordinator: Dr. Sumana M	

Course Content

Unit I

The Object Oriented Paradigm – Functional Decomposition, The Problem with Requirements, Dealing with Changes, The OO Paradigm; **The UML—The Unified Modeling Language:** Overview, What Is the UML? Why Use the UML?, The Class Diagram, Interaction Diagrams; The Principles and Strategies of Design Patterns: The Open-Closed principle, Designing from Context, Encapsulating Variation, Abstract classes vs Interfaces, The principle of Healthy Skepticism

Recommended Resources / Links:

- <https://www.udemy.com/course/design-patterns-java/>
- <https://nptel.ac.in/courses/106105224>

Unit II

An Introduction to Design Patterns: Overview, Design Patterns Arose from Architecture and Anthropology, Moving from Architectural to Software Design Patterns, Why Study Design Patterns?, Other Advantages to Studying Design Patterns, Summary; **The Facade Pattern:** Overview, Introducing the Façade Pattern, Learning the Façade Pattern, Field Notes: The Façade Pattern; **The Adapter Pattern:** Overview, Introducing the Adapter Pattern, Learning the Adapter Pattern, Field Notes: The Adapter Pattern; **Expanding Our Horizons** – Objects: Traditional Vs New Views, Encapsulation: Traditional Vs New Views, Finding Varying Concept and Encapsulating, Commonality and Variability Analysis and Abstraction

Recommended Resources / Links:

- <https://www.udemy.com/course/design-patterns-java/>
- <https://nptel.ac.in/courses/106105224>

Unit III

The Strategy Pattern: Overview, The International E-Commerce System Case Study: Initial Requirements, Handling New Requirements, The Strategy Pattern; **The Bridge Pattern:** Overview, Introducing the Bridge Pattern, Learning the Bridge Pattern – An example, An Observation About Using Design Patterns, Learning the Bridge Pattern – Deriving It, The Bridge Pattern in retrospect; **The Abstract Factory Pattern:** Overview, Introducing the Abstract Factory Pattern, Learning the Abstract Factory Pattern – An example, Learning the Abstract Factory Pattern – Implementing It

Recommended Resources / Links:

- <https://www.udemy.com/course/design-patterns-java/>
- <https://nptel.ac.in/courses/106105224>

Unit IV

The Observer Pattern: Overview, Categories of Patterns, More Requirements for the International E-Commerce Case Study, The Observer Pattern, Applying the Observer to the Case Study; The Template Method Pattern: Overview, More Requirements for the International E-Commerce Case Study, The Template Method Pattern, Applying the Template Method to the International E-Commerce Case Study, Using the Template Method Pattern to Reduce Redundancy

Recommended Resources / Links:

- <https://www.udemy.com/course/design-patterns-java/>
- <https://nptel.ac.in/courses/106105224>

Unit V

Lessons from Design Patterns: Factories – Factories, The Universal Context Revisited, Factories Follow our Guidelines; **The Singleton Pattern and the Double-Checked Locking Pattern:** Overview, Introducing the Singleton Pattern, Applying the Singleton Pattern to the Case Study; **A Variant:** The Double-Checked Locking Pattern, Reflections, Use the Singleton and Double-Checked Locking Patterns if applicable for a given problem; **The Factory Method Pattern:** Overview, More Requirements for the Case Study, The Factory Method Pattern, Factory Method Pattern and Object-Oriented Languages

Recommended Resources / Links:

- <https://www.udemy.com/course/design-patterns-java/>
- <https://nptel.ac.in/courses/106105224>

Text Books:

1. Alan Shalloway, James R Trot, “Design Patterns Explained – A New Perspective on Object-Oriented Design”, Pearson, 2nd Edition, 4th Impression 2010
2. Eric Freeman, Elisabeth Freeman, “Head First Design Patterns”, O’reilly Publications, October 2004, 1st Edition
3. Satzinger, Jackson, Burd, “Object Oriented Analysis and Design with Unified Process”, Thomson Learning, 1st Indian Reprint 2007.

Activity Based Learning (Suggested Activities in Class)/ Practical Based learning

- Case Study Discussion
- Programming assignment

Course Outcomes (COs):

At the end of the course, the student will be able to:

1. Design solution to the given problem definition using UML notations (PO – 1, 2, 3) (PSO – 1)
2. Apply Façade and Adapter patterns to given problem with traditional design (PO – 1, 2, 3, 9, 10, 12) (PSO – 1)
3. Apply Strategy, Bridge and Abstract Factory patterns to the given scenario (PO – 1, 2, 3, 9, 10, 12) (PSO – 1)
4. Apply Template Method and/or Observer patterns for a given problem design (PO – 1, 2, 3, 9, 10, 12) (PSO – 1)
5. Use Singleton, Double checked locking pattern and Factory method pattern for a given problem design (PO – 1, 2, 3, 9, 10, 12) (PSO – 1)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
Assignment	10	CO1, CO2, CO3, CO4, CO5
Case based Design evaluation	10	CO1, CO2, CO3, CO4, CO5
Semester-End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5

RESEARCH METHODOLOGY AND INTELLECTUAL PROPERTY RIGHTS	
Course Code: AL58	Credits : 3:0:0
Prerequisite: Nil	Contact Hours: 42

Course Content

Unit I

Research Methodology

Introduction: Meaning of Research, Objectives of Research, Types of Research, Ethics in Research, Types of Research Misconduct. Literature Review and Technical Reading, New and Existing Knowledge, Analysis and Synthesis of Prior Art, Bibliographic Databases, Conceptualizing Research, Critical and Creative Reading. **Citations:** Functions and Attributes, Impact of Title and Keywords on Citations, Knowledge flow through Citations, Acknowledgments, and Attributions.

- Pedagogy: Chalk and Talk, PowerPoint Presentations
- Links: https://onlinecourses.nptel.ac.in/noc22_ge08/preview

Unit II

Research Design: Need for Research Design, Important Concepts Related to Research Design: Dependent and Independent Variables, Extraneous Variable, Variable, Common Control, Confounded Relationship, Research Hypothesis, Experimental and Control Groups, Treatments.

Experimental Designs: Introduction to Randomized Block Design, Complete Randomized Design, Latin Square Design, and Factorial Design.

- Pedagogy: Chalk and Talk, PowerPoint Presentations
- Links: https://onlinecourses.nptel.ac.in/noc22_ge08/preview

Unit III

Method of Data Collection: Primary and Secondary Data Collection.

Sampling Design: Sampling fundamentals, Measurement, and Scaling Techniques, Criteria of Selecting a Sampling Procedure, Characteristics of a Good Sample Design, and Types of Sample Design. **Data Analysis:** Testing of Hypotheses: Null Hypothesis, Alternative Hypothesis, Type I and Type II Errors, Level of Significance. Procedure for Hypothesis Testing: Mean, Variance, Proportions. Chi-square Test, Analysis of Variance (One Way ANOVA), and Covariance (ANOCOVA)

- Pedagogy: Chalk and Talk, PowerPoint Presentations
- Links: https://onlinecourses.nptel.ac.in/noc23_ge36/preview

Unit IV

Intellectual Property Rights

Introduction to IPR: Different forms of IPR, Role of IPR in Research and Development. TRIPS Agreement, Patent Cooperation Treaty (PCT). **Patents:** Brief history of Patents-Indian and Global Scenario, Principles Underlying Patent Law, Types of Patent Applications in India, Procedure for

Obtaining a Patent. Non Patentable Inventions. Rights Conferred to a Patentee, Basmati Rice Patent Case.

- Pedagogy: Chalk and Talk, PowerPoint Presentations
- Links: <https://archive.nptel.ac.in/courses/110/105/110105139/>

Unit V

Design: What is a Design? Essential Requirements for a Registrable Design, Procedure of Registration of a Design, **Trademarks:** Essentials of a Trademark, Registration, and Protection of Trademarks, Rights Conferred by Registration of Trademarks, Infringements, Types of Reliefs, Case Studies. **Copyrights:** Characteristics of Copyrights, Rights Conferred by Registration of Copyrights, Registration of Copyrights, Infringements, Remedies against Infringement of Copyrights, Case studies

- Pedagogy: Chalk and Talk, PowerPoint Presentations
- Links: <https://archive.nptel.ac.in/courses/110/105/110105139/>

Text Books:

1. C. R Kothari, Gourav Garg, Research Methodology – Methods and Techniques. New Age International Publishers.
2. Dr. B L Wadehra – Law relating to Intellectual property. Universal Law Publishing Co.
3. Dipankar Deb, Rajeeb Dey, Valentina E. Balas “Engineering Research Methodology”, ISSN 1868-4394 ISSN 1868-4408 (electronic), Intelligent Systems Reference Library, ISBN 978-981-13-2946-3 ISBN 978-981-13-2947-0 (eBook), <https://doi.org/10.1007/978-981-13-2947-0>.

Reference Books:

1. David V. Thiel “Research Methods for Engineers” Cambridge University Press, 978-1-107-03488-4

Course Outcomes (COs):

At the end of the course, the student will be able to:

1. Possess the knowledge of research and conduct a literature review. (PO-8, PO-10, PO-12)
2. Apply the knowledge of research design and design of experiments. (PO-4, PO-8, PO 10, PO-12)
3. Analyze data collection methods, analysis, and sampling design. (PO-4, PO-8, PO-10, PO-12)
4. Understand the global and Indian scenarios of patents and patent applications. (PO-8, PO-10, PO-12)
5. Acquire the requirements of registration and infringements related to trademarks, copyrights, and designs. (PO-8, PO-10, PO-12)

NOSQL DATABASE	
Course Code: ISAECS591	Credits: 1:0:0
Pre – requisites: NIL	Contact Hours: 14
Course Coordinator: Dr. Lincy Mathews	

Course Contents

Unit 1

Variety of NoSQL Databases: Data Management with Distributed Databases, ACID and BASE, Four Types of NoSQL Databases.

Pedagogy: PPT, Chalk and Talk Method

Links:

- https://youtu.be/F4CzPATviaE?si=p9nZ79yvLICzBG_Y
- https://www.coursera.org/learn/introduction-to-nosql-databases?utm_source=chatgpt.com

Unit 2

Key Value Databases: From Arrays to Key Value Databases, Keys: More than Meaningless Identifiers, Key Value Database Terminology: Key Value Database Data Modelling Terms, Key Value Architecture terms, Designing for Key value, Databases: Key Design and Partitioning

Pedagogy: PPT, Chalk and Talk Method

Links:

- <https://youtu.be/-s29jUAkg70?si=lbtA5wVJz5hqGmsW>
- <https://youtu.be/G7t-UKWZPr0?si=0gsbm1bg5rp38rRk>

Unit 3

Document Databases: Introduction to Document Databases: Basic Operations on Document Databases, Document Database terminology: Document and Collection Terms, Types of Partition, Data Modelling and Query Processing

Pedagogy: PPT, Chalk and Talk Method

Links:

- <https://www.youtube.com/watch?v=mMdZEAKtntA>

Unit 4

Column Family Databases: Introduction, Differences and Similarities to Key-value and Document Databases, Column Family Database Terminology: Basic Components, Structures and Processes, Designing for Column Family Databases: Guidelines for Designing Tables, Guidelines for Indexing

Pedagogy: PPT, Chalk and Talk Method

Links:

- <https://www.youtube.com/watch?v=Nh6Y7DgZDrg>

Unit 5

Graph Databases: Introduction to Graph Databases, Graphs and network Modelling, Advantages of Graph databases, Graph database Terminology: Elements of graphs, operations on Graphs, Properties, types of Graphs, Designing for Graph databases : Querying a graph.

Pedagogy: PPT, Chalk and Talk Method

Links:

- <https://www.youtube.com/watch?v=VfcRxtBKI54>

Text Books:

Dan Sullivan, NoSQL for Mere Mortals, Pearson publication, 2016 Edition

References:

Sadalage, P. & Fowler, NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence, Pearson Addison Wesley, 2012

Self-Learning (Programming Assignment/ Case Study/Project-Based Learning/MOOC-based learning / GATE-oriented aptitude test / Simulation):

Course Outcomes:

1. Explain the emergence and different types of No SQL Databases. (PO-1,2 ,7,9,10,11,12 & PSO-2)
2. Understand the modelling concepts for a Key Value Database. (PO-1,2,7,9,10,11,12 & PSO-2)
3. Demonstrate the detailed architecture of Document Databases (PO-1,2,5,7,9,10,11,12 & PSO-2).
4. Demonstrate the detailed architecture of Column Family Databases (PO-1,2,5,7,9,10,11,12 & PSO-2).
5. Demonstrate the detailed architecture of Graph Databases. (PO-1,2,5,7,9,10,11,12 & PSO-2).

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2, CO3
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
Case Study	20	CO1, CO2, CO3, CO4, CO5
Semester-End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5

FRONTEND DEVELOPMENT USING REACT JS	
Course Code: ISAECS594	Credits:1:0:0
Pre – requisites: NIL	Contact Hours:14
Course Coordinator: Shruthi J R	

Course Content

UNIT 1

Introduction to React and JSX: React Philosophy: Declarative, Component-Based, React vs. Other Frameworks (Angular, Vue), Introduction to JSX: JSX Syntax, Embedding Expressions, Conditional Rendering in JSX, JSX Compilation with Babel & JSX Transform

Links: <https://legacy.reactjs.org/docs/introducing-jsx.html>
<https://reactnative.dev/docs/next/intro-react>

UNIT 2

Components and Props: Components: Functional vs. Class, Creating and Rendering Components, Passing & Accessing Props, Difference between Props and State, this.props.children and React Fragments, Component Lifecycle (Overview), Rendering Components

Links: <https://reactnative.dev/docs/next/intro-react>
<https://legacy.reactjs.org/docs/components-and-props.html>
<https://legacy.reactjs.org/docs/react-component.html>

UNIT 3

Events and Form Handling: Handling Events in React: SyntheticEvent, Arrow Functions, bind(), Event Handling in Functional and Class Components, Controlled Inputs (Forms), Form Elements: input, textarea, select, Lifting State Up

Links: <https://legacy.reactjs.org/docs/handling-events.html>
<https://www.freecodecamp.org/news/how-to-handle-events-in-react-19/>
<https://www.scaler.com/topics/react/event-handling-in-react/>
<https://www.angularminds.com/blog/best-practices-of-form-handling-and-event-binding-in-react>

UNIT 4

React DevTools and Routing: Introduction to React Developer Tools: Inspecting Components, Props, State, Introduction to React Router: Setting up Routes, Navigating with Link and NavLink, Using useParams and useLocation

Links: https://www.youtube.com/watch?v=943D7U74_sQ
<https://www.contentful.com/blog/react-routing/>
<https://www.geeksforgeeks.org/reactjs/link-and-navlink-components-in-react-router-dom/>
<https://blog.logrocket.com/react-router-dom-tutorial-examples/>

UNIT 5

Styling, Deployment Basics & Project Integration: Styling in React: CSS Modules, Inline Styles, CSS-in-JS, Overview of Hooks (useEffect, useState, useRef), .env setup, Basic Deployment Concepts: Build Process, Netlify Hosting Overview

Links: <https://www.freecodecamp.org/news/style-react-apps-with-css/>
<https://thehumblecoder.medium.com/mastering-css-styling-in-react-js-a-practical-guide-697de390f215>
<https://docs.netlify.com/build/frameworks/framework-setup-guides/react/>

Suggested Learning Resources:

Text Books:

1. Chris Minnick, Beginning ReactJS Foundations Building User Interfaces with ReactJS An Approachable Guide, 2022.
2. Anthony Accomazzo, Ari Lerner, Nate Murray, Clay Allsopp, David Gutman, and Tyler McGinnis, Fullstack React The Complete Guide to ReactJS and Friends, 2017.

Course outcomes

At the end of the course, the student will be able to:

- 1 Understand the core principles of React including JSX syntax, component-based architecture, and its comparison with other frameworks. (PO-1, PO-2, PO-12, PSO-1)
2. Apply knowledge of functional and class components, props and state management to build dynamic user interfaces. (PO-1, PO-2, PO-3, PO-5, PSO-1, PSO-2)
3. Develop and handle events and forms using controlled components and implement component-level state lifting to manage data flow in React applications. (PO-1, PO-2, PO-3, PO-10, PSO-1, PSO-2)
4. Utilize React Developer Tools and React Router to inspect application state, manage navigation, and build single-page applications. (PO-1, PO-3, PO-5, PO-11, PSO-1, PSO-2)
5. Apply styling techniques, React Hooks like useEffect, and deployment methods to build and host complete front-end projects. (PO-1, PO-3, PO-5, PO-12, PSO-1, PSO-2, PSO-3)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2, CO3
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
Learning Project	20	CO1, CO2, CO3, CO4, CO5
Semester-End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5

ENVIRONMENTAL STUDIES	
Course Code: HS510	Credits : 0:0:0
Pre - requisite: Nil	Contact Hours: 14L

Course Content

Unit I

Environment, Ecology and Biodiversity

Definition, scope and importance. Multidisciplinary nature of Environmental studies. Food chain and food web. Energy flow and material cycling in ecosystem. Biodiversity and threats to biodiversity. Concept of sustainable development: Definition, objectives and applications.

- Pedagogy/Course delivery tools: Chalk and Talk, Power point presentations, Videos, Models
- Link: https://youtu.be/I_bnGkviWOU
- Link: <https://youtu.be/Ar04qG1P8Es>

Unit II

Natural resources

Forest resources: Ecological importance of forests. Water resources: Global water resources distribution. Mineral resources: Environmental effects of extracting and processing Mineral resources. Food resources: Effects of modern agriculture. Land resources: Soil erosion and Desertification.

- Pedagogy/Course delivery tools: Chalk and Talk, Power point presentations, Videos
- Link: <https://youtu.be/vsXv3anIBSU>
- Link: <https://youtu.be/1rOVPqaUyv8>

Unit III

Energy sources

Growing energy needs. Conventional and non-conventional / Renewable and Non-renewable energy sources. Bio Energy-Ethanol and Bio mass energy. Energy of the future – Hydrogen fuel cells and Nuclear energy. Environmental Impact Assessment (EIA): Definition, Objectives and benefits. Step by step procedure of conducting EIA.

- Pedagogy/Course delivery tools: Chalk and Talk, Power point presentations, Animations, Models
- Link: <https://youtu.be/mh51mAUexK4>
- Link: https://youtu.be/XS-eXqppf_w

Unit IV

Environmental pollution

Definition, Causes, Effects and control measures of Water pollution, Air pollution and Soil/ land pollution. Management of Municipal Solid Waste and treatment methods of municipal solid waste.

- Pedagogy/Course delivery tools: Chalk and Talk, Power point presentations, Videos
- Link: <https://youtu.be/NRoFvz8Ugeo>

- Link: <https://youtu.be/DAQapF-F4Vw>

Unit V

Environmental protection

Global warming and Climate change, Acid rain, Ozone layer depletion. Salient features of Environmental Protection Act, Air & Water Acts. Functions of Central and State Pollution Control Boards.

- Pedagogy/Course delivery tools: Chalk and Talk, Power point presentations, Open source software
- Link: <https://youtu.be/iV-BvYwl4Y8>
- Link: <https://youtu.be/BYqLRGawoH0>

Text Books:

1. **Dr. S M Prakash** – Environmental Studies, Elite Publishers, 2007.

Reference Books:

1. **P. Venugopala Rao** – Principles of Environmental Science & Engineering Prentice Hall of India, 1st edition, 2006.

Web links and video Lectures (e- Resources):

1. https://youtu.be/I_bnGkviWOU
2. <https://youtu.be/vsXv3anIBSU>
3. <https://youtu.be/mh51mAUexK4>
4. <https://youtu.be/NRoFvz8Ugeo>
5. <https://youtu.be/iV-BvYwl4Y8>

Course Outcomes (COs):

At the end of the course, the student will be able to:

1. Describe the importance of environmental studies, sustainable development and biodiversity (PO-1, 7)
2. Explain the importance and conservation of impacts of natural resources (PO-1, 7)
3. Distinguish the energy sources and identify the alternative energy sources for sustainable development (PO-1, 7)
4. Identify the causes, effects and control measures of pollution in developmental activities (PO-1, 7)
5. Outline the current environmental issues and the role of the agencies for environmental protection (PO-1, 7)

VI SEMESTER

MANAGEMENT & ENTREPRENEURSHIP	
Subject Code: AL61	Credits: 3:0:0
Pre requisites: NIL	Contact Hours: 42L
Course Coordinators: Dr. M Shilpa/ Dr. M Rajesh	

Course Content

Unit I

Introduction to Management: Definition of Management, Its nature and purpose, Contributions of F.W. Taylor and Henry Fayol to management theory, Functions of managers. **Planning:** Types of plans, Steps in planning, the planning process, Management by Objectives (MBO) **Organizing:** The nature and purpose of organizing, Formal and informal organization. Organization levels and Span of management, Principle of span of management, the structure and process of organizing

- Pedagogy: Chalk and Talk, PowerPoint Presentations
- Links: https://onlinecourses.nptel.ac.in/noc23_mg33/preview
<https://www.digimat.in/nptel/courses/video/110107150/L01.html>

Unit II

Staffing: Situational factors affecting staffing. **Leading:** Human factors in managing, definition of leadership, Ingredients of leadership **Controlling:** Basic control process, Critical control points and standards, Control as a feedback system, Feed forward control, Requirements for effective controls.

- Pedagogy: Chalk and Talk, PowerPoint Presentations
- Links: <https://nptel.ac.in/courses/110107150>

Unit III

Introduction to Entrepreneurship: The Foundations of Entrepreneurship: What is an Entrepreneurship? The benefits of Entrepreneurship, The potential drawbacks of Entrepreneurship; Inside the Entrepreneurial Mind: From Ideas to Reality: Creativity, Innovation and Entrepreneurship, Creative Thinking, Barriers to Creativity

- Pedagogy: Chalk and Talk, PowerPoint Presentations
- Links: <https://www.youtube.com/watch?v=HgkRrvbhQ&list=PL7oBzLzHZ1wXW3mtolxV5nIGn48NLKwrb>

Unit IV

The Entrepreneurial Journey: Crafting a Business Plan: The benefits of creating a business plan, the elements of a business plan; Forms of Business Ownership and Buying an Existing Business: Sole proprietorships and partnership. **Launching the Business:** Franchising and the Entrepreneur: Types of Franchising, the benefits of buying a Franchise

- Pedagogy: Chalk and Talk, PowerPoint Presentations
- Links: <https://www.youtube.com/watch?v=Tzzfd6168jk&list=PLyqSpQzTE6M8EGZbmNUuUM7Vh2GkdbB1R>
https://www.youtube.com/watch?v=5RMqxtMwejM&list=PLyqSpQzTE6M9zMKj_PSm81k9U8NjaVJkR

Unit V

Small Scale Industry: Definition, Characteristics, types, role of SSI in Economic Development, steps to start SSI- Govt. Policy towards SSI, Institutional support- TECKSOK, KIADB, KSSIDC, KSIMC, DIC, NSIC, SIDBI, KSFC.

- Pedagogy: Chalk and Talk, PowerPoint Presentations
- Links: <https://www.nptel.ac.in/courses/110/105/110105067/>

Text Books:

1. Harold Koontz, H. Weihrich, and A.R. Aryasri, Principles of Management, Tata McGraw-Hill, New Delhi, 2004.
2. Essentials of Entrepreneurship and Small Business Management – Norman Scarborough & Jeffrey Cornwall (Pearson, 2016)
3. Management and Entrepreneurship by K. R. Phaneesh, Sudha Publications, Bangalore

References:

1. Innovation & Entrepreneurship – Peter Drucker (Harper, 2006)
2. Entrepreneurship: The Art, Science, and Process for Success – Charles Bamford & Garry Bruton (McGraw-Hill, 2015)
3. Management and Entrepreneurship-NVR Naidu, T Krishna Rao, I.K. International Publishing House Pvt. Ltd.@ 2008
4. Poornima M Charantimath, Entrepreneurship Development and Small Business Enterprises, Pearson Education, 2006.

Course Outcomes (COs):

At the end of the course, student will be able to

1. Plan and organize for the manpower in the given type of organization (PO-6,9,11)
2. Use staffing Leading and controlling functions for the given organization. (PO-6,8,9,10)
3. Analyze the advantages and potential drawbacks of Entrepreneurship. (PO-6,7,8)
4. Develop a basic business plan and analyze the benefits of buying Franchise. (PO-3,5,7,8,11)
5. Identify the various institutions that provide financial support to small scale industries. (PO-6,11)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment Tool	Marks	Course outcomes addressed
Internal test-I	30	CO1, CO2
Internal test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30marks.		
Other components		
Quiz	10	CO1, CO2, CO3
Assignment	10	CO1, CO2, CO3, CO4, CO5
Semester End Examination (SEE)	100	CO1, CO2, CO3, CO4, CO5

MACHINE LEARNING	
Course Code: IS62	Credits: 3:1:0
Pre – requisites: Engg. Mathematics	Contact Hours: 42+14 hrs.
Course Coordinator: Dr. Savita K Shetty	

Course Contents

Unit 1

Machine Learning Introduction: Learning, Types of Machine Learning, Supervised Learning, The Machine Learning Process.

Machine Learning Preliminaries: Terminology - Weight Space, The Curse of Dimensionality; Testing Machine Learning Algorithms – Over-fitting, Training, Testing and Validation Sets, The Confusion Matrix, Accuracy Metrics, ROC Curve, Unbalanced Dataset, Measuring Precision.

Turning Data into Probabilities: Minimizing Risk, maximum a posteriori hypothesis; Basic Statistics: Averages, Variance and Covariance, The Gaussian; Bias-Variance Trade-off.

Recommended Resources / Links:

1. https://www.youtube.com/watch?v=T3PsRW6wZSY&list=PLIg1dOXc_acbdJo-AE5RXpIM_rvwrerwR

Unit 2

Measuring Data Similarity and Dissimilarity: Data Matrix versus Dissimilarity Matrix, Proximity Measures for Nominal Attributes, Proximity Measures for Binary Attributes, Dissimilarity of Numeric Data: Minkowski Distance, Proximity Measures for Ordinal Attributes, Dissimilarity for Attributes of Mixed Types, Cosine Similarity.

Concept Learning: Concept learning task, Concept learning as search, Find-S algorithm, Version space, Candidate Elimination algorithm, Inductive Bias. Problems on Candidate Elimination and Find-S.

Dimensionality Reduction - Supervised: LDA (Linear Discriminant Analysis) -Technique, Examples as Numerical.

Recommended Resources / Links: <https://www.youtube.com/watch?v=z5AKsT3apWI>
<https://www.youtube.com/watch?v=gwtTM9M6ugU&t=10s>

Unit 3

Supervised Learning Algorithms - Regression: Linear Regression - Technique, Examples as Numerical; Multiple Linear Regression, Logistic Regression Examples as Numerical, ID3, CART, K-Nearest Neighbour Methods - Technique, Examples as Numerical

Probabilistic Learning: The Naive Bayes Classifier -examples and numerical

Support Vector Machines: Linear and Nonlinear - Technique, Examples as Numerical;

Recommended Resources / Links:

https://www.youtube.com/watch?v=8PJ24SrQy8&list=PLIg1dOXc_acbdJo-AE5RXpIM_rvwrerwR&index=6
https://www.youtube.com/watch?v=FuJVLsZyKuE&list=PLIg1dOXc_acbdJo-AE5RXpIM_rvwrerwR&index=7

Unit 4

Unsupervised Learning: Partitional Clustering - K-Means, Dealing with noise, Examples as Numerical, Elbow method to choose the right value of 'k', problems with k-means clustering,

Hierarchical Clustering - Agglomerative(AGNES), Divisive(DIANA), Examples as numericals, Density-based (DBSCAN: Density-Based Clustering Based on Connected Regions with High Density), Expectation maximization.

Recommended Resources / Links:

https://www.youtube.com/watch?v=qg_M37WGKG8&list=PLIg1dOXc_acbdJo-AE5RXpIM_rvwrerwR&index=42

https://www.youtube.com/watch?v=NCsHRMkDRE4&list=PLIg1dOXc_acbdJo-AE5RXpIM_rvwrerwR&index=43

Unit 5

Dimensionality Reduction - Unsupervised: Introduction, Subset Selection, PCA (Principal Component Analysis) – Technique, Examples as Numerical.

Model selection & evaluation: Holdout Method and Random Subsampling, Cross-Validation, Bootstrap, Model Selection Using Statistical Tests of Significance.

Introducing Ensemble Methods: Bagging - Random forests, Boosting -AdaBoost, XGBoost
Examples as Numerical

Recommended Resources / Links:

https://www.youtube.com/watch?v=nYCAH8b5AQ0&list=PLIg1dOXc_acbdJo-AE5RXpIM_rvwrerwR&index=4

https://www.youtube.com/watch?v=nelJ3svz0_o&list=PLIg1dOXc_acbdJo-AE5RXpIM_rvwrerwR&index=39

TUTORIALS

- 1) Confusion Matrix and Accuracy Metrics, Proximity Measures for Nominal, Binary, Numeric, Ordinal Attributes
- 2) Find-S and Candidate elimination algorithm
- 3) LDA (Linear discriminant analysis)
- 4) Linear Regression and Logistic Regression
- 5) ID3 – Classification
- 6) CART – Classification and regression
- 7) K-Nearest Neighbor Classifier
- 8) Naive Bayes Classifier
- 9) Support vector Machine-Linear and Non Linear
- 10) Clustering: K-Means, AGNES, DIANA, DBCAN
- 11) PCA (Principal Component Analysis)
- 12) Ensemble methods (Random Forest, AdaBoost, XGBoost)

Text Books:

1. Stephen Marsland, “Machine Learning - An Algorithmic Perspective”, Second Edition, CRC Press - Taylor and Francis Group, 2015
2. S Sridhar and M Vijayalakshmi, “Machine Learning” Oxford University Press, 2021

Reference Books:

1. Tom Mitchell, —Machine Learning, McGraw Hill, 3rd Edition, 1997.
2. Jiawei Han, Micheline Kamber and Jian Pei, Data Mining: Concepts and Techniques, 4th ed, The Morgan Kaufmann Series in Data Management Systems Morgan Kaufmann Publishers, July 2011.
3. Ethem Alpaydin, “Introduction to Machine Learning”, 4th Edition, MIT Press, Prentice Hall of India (PHI) Learning Pvt. Ltd. 2020

Course Outcomes:

At the end of the course, the student will be able to:

1. Understand various types of data and their preprocessing methods. (PO- 1) (PSO- 3)
2. Analyze the various performance metrics used in machine learning. (PO- 1, 2, 9, 10) (PSO- 3)
3. Use supervised learning methods for Classification and Regression (PO- 1, 2, 3, 9, 10, 12) (PSO- 3)
4. Apply appropriate clustering techniques for a given scenario (PO- 1, 2, 3, 9, 10, 12) (PSO- 3)
5. Understand various validation methods for appropriate model selection (PO- 1) (PSO- 3)

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2, CO3
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
Tutorial Evaluation	10	CO1, CO2, CO3, CO4, CO5
Test	10	CO1, CO2, CO3, CO4, CO5
Semester-End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5

MACHINE LEARNING LAB	
Course Code: ISL65	Credits: 0:0:1
Pre – requisites: Python	Contact Hours: 28
Course Coordinator: Mrs. Kavya K S	

List of Experiments

Part-A

- Using **any dataset**, calculate TP, TN, FP ,FN and different metrics (Accuracy, Precision, Recall(Sensitivity), F1-Score, MCC, Specificity, Negative Predictive Value) by defining your own functions. Compare your values with scikit-learn's library functions. Get the result of Confusion Matrix using sklearn. Using sklearn, plot the ROC & AUC Curves for your test data and random probabilities. Using sklearn, calculate the AUC of your test data and of random probabilities. Interpret the results. Write the inference/analysis of each output.
- Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a set of training data samples. Create your own dataset with minimum 4 attributes.
- Implement linear and logistic regression and test it using any dataset. Give new test data and predict the output. Print the confusion matrix, accuracy, precision, recall, MSE , RMSE etc. Compare and write the inference.
- Write a program to demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample. Interpret the results. Write the inference/analysis of each output.
- Build a KNN model for predicting if a person will have diabetes or not with a high accuracy score. Perform some appropriate Pre-Processing steps on the given dataset for better results. Implement the KNN algorithm on your own. Try other possible processes that can be done to dataset and tuning the model to increase accuracy such as Increase K value, Normalization and Different Distance Metrics
- Create a dataset and Perform the necessary pre-processing steps. Train the model using Naive Bayes Classifier. Give new test data and predict the classification output. Analyze and write the inference.
- Build a K-Means Model for any dataset. Assume K value as 2,3,4 .Compare and interpret the results of different clusters.
- Using any dataset implement Hierarchical Clustering (AGNES and DIANA). Plot the Dendrogram for Hierarchical Clustering and analyze your result. Plot the clustering output for the same dataset using these two hierarchical techniques. Compare the results. Write the inference.
- Write a program to implement **DBSCAN** on any dataset. Apply DBSCAN with different values of **eps** and **min_samples**. Visualize the resulting clusters and noise points. Write the inference/analysis of each output.
- Write a program to demonstrate **PCA** on any high-dimensional dataset. Perform PCA to reduce the dataset to **2 principal components**. Visualize the transformed data in a 2D scatter plot. Write the inference/analysis of each output.
- Design an experiment to compare the performance of AdaBoost and XGBoost algorithms on a binary classification task using a dataset of your choice. Write Python code to implement both algorithms and evaluate their performance using appropriate evaluation metrics.
- Design an experiment to investigate the impact of varying the number of trees in a Random Forest classifier on its performance for a given dataset. Write Python code to implement the

Random Forest algorithm with different numbers of trees and evaluate its classification performance using appropriate evaluation metrics.

Course outcomes:

At the end of the course the student will be able to:

1. Understand and implement various machine learning procedures algorithms using appropriate datasets. (PO: 1, 2, 3, 4, 5, PSO: 1, 3)
2. Design and develop Java/Python programs for supervised, unsupervised, and ensemble learning methods. (PO: 1, 2, 3, 4, 5, PSO: 1, 3)
3. Apply machine learning techniques to analyze and solve real-world problems. (PO: 1, 2, 3, 4, 5, PSO: 1, 3)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test	20	CO1, CO2, CO3, CO4, CO5
Other Components		
Write-up, Execution, Record, Viva, Project and project report	30	CO1, CO2, CO3, CO4, CO5
Semester-End Examination (SEE)	50	CO1, CO2, CO3, CO4, CO5

DEVOPS LAB	
Course Code: 22ISL66	Credits: 0:0:1
Pre – requisites:	Contact Hours: 28
Course Coordinator: Ms. J R Shruti	

Course Contents

List of Experiments

1. **Project Identification and Module Design:** Identify a project with distinct modules. Prepare a project plan with detailed architecture, technologies to be used, and responsibilities.
2. **Project Development and Modular Implementation:** Build the project modules with clean code structure, appropriate naming conventions, and folder hierarchies.
3. **Version Control Basics using Git:** Practice Git operations: init, add, commit, status, log, diff, and working with local repositories.
4. **Remote Git Repository and Collaboration with GitHub:** Use GitHub for remote version control: clone, push, pull, merge, stash, branch management, and resolving merge conflicts.
5. **Agile Development and Backlog Management:** Use Kanban boards, sprints, and task assignment tools (e.g., Azure Boards, GitLab Issues, or Jira) to manage project progress.
6. **Dashboard and Reporting Tools for Project Tracking :** Create dashboards and reporting tools to visualize work progress and productivity. Integrate project boards with Git or CI tools if applicable.
7. **Continuous Integration Setup using Jenkins or GitLab CI/CD:** Configure Jenkins or GitLab CI/CD pipelines to automate code building and testing upon push events.
8. **Deployment Automation Using CI/CD Tools:** Extend CI to CD: auto-deploy project to a staging/production environment using Azure Pipelines, Jenkins, or similar.
9. **Infrastructure Automation with Configuration Management Tools:** Use Ansible, Puppet, or Chef to write playbooks or manifests to configure and provision environments.
10. **Containerization using Docker:** Write a Dockerfile and containerize the application. Run and manage containers with Docker CLI.
11. **Container Orchestration using Kubernetes:** containerized project into Kubernetes. Use kubectl to create pods, services, deployments, etc.
12. **Test Automation for the Project :** Write and run automated test cases using tools like Selenium, JUnit, PyTest, or TestNG. Integrate with CI pipeline if possible.

Text Books:

1. Aashikha Narasimhan, Roshan Sham, Harish Subramanian, A Beginner's Handbook To Devops, 2023.
2. Mikael Krief, Learning DevOps- A comprehensive guide to accelerating DevOps culture adoption with Terraform, Azure DevOps, Kubernetes, and Jenkins, Second Edition, 2022.
3. Jennifer Davis and Ryn Daniels, Effective DevOps Building a Culture of Collaboration, and Tooling at Scale, 2016.

Reference Books:

1. Milindanath Hewage, A Practical Guide to Azure DevOps Learn by doing Third Edition, 2016.

Course Outcomes:

At the end of the course, the student will be able to:

1. Demonstrate the DevOps culture by illustrating application's cloud infrastructure and configuration management with Ansible. (PO-2, 3, 5, 9, PSO-3)
2. Apply the DevOps pipeline process starting with continuous integration and continuous deployment principles. (PO-2, 3, 5, 9, PSO-3)
3. Demonstrate how to create and run a container from a Docker file and deploy a complex application on Kubernetes. (PO-2, 3, 5, 9, PSO-3)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Continuous Evaluation and Quiz	20	CO1, CO2, CO3
Other Components		
Tool Demo for various stages of DevOps using Jenkins/Azure DevOps, Record and Viva	30	CO1, CO2, CO3
Semester-End Examination (SEE) Write-Up: 10 Marks Tool Demo for various stages of DevOps using Jenkins/Azure DevOps a. Version Control: 5 Marks b. Setting the pipeline for various stages(groovy /yaml code)- 15 Marks c. Containerization: 5 Marks d. Deployment: 5 Marks Viva-voce: 10 Marks	50	CO1, CO2, CO3

MOBILE APPLICATION DEVELOPMENT-2	
Course Code: ISE631	Credits: 3:0:0
Pre – requisites: Mobile Application Development-1	Contact Hours: 42
Course Coordinator: Dr. S R Mani Sekhar	

Course Contents

Unit 1

Scroll View and Table View Controllers

Protocols - CustomStringConvertible, Equatable, Comparable, Codable, Creating Protocol, Delegation, App Anatomy and Life-Cycle - Stages of App Life-Cycle, AppDelegate Methods, SceneDelegate Methods, Example code - App Event Count, Model-View-Controller - Meal List TableView Controller, Project Organisation, Example code - Favorite Athletes, Scroll Views - Example code - I Spy, Table Views - Anatomy of a TableView, Custom TableView Cells, Edit TableViews,

Recommended Resources / Links:

- <https://books.apple.com/us/book/develop-in-swift-data-collections/id1581183203>
- <https://www.youtube.com/@iOSAcademy>

Unit 2

System view controllers and data persistence. Saving Data - Encoding and Decoding using Codable, Writing Data to a File, System View Controllers - Share with the Activity Controller, Use Safari Services to Display Web Contents, present an alertController, Access the Camera, Send Email from your App, Example Code - Home Furniture Sharing, Complex Input Screens – input screen, collect dates, Date Pickers- show or hide, Binary Input.

Recommended Resources / Links:

- <https://books.apple.com/us/book/develop-in-swift-data-collections/id1581183203>
- <https://www.youtube.com/@iOSAcademy>

Unit 3

Closures - Syntax, Passing Closures as Arguments, Additional Syntactic Sugars while using Closures as Parameters, Collection Functions using Closures (map, reduce, filter and sort(by:)) , Extensions - Adding Computed Properties, Adding Instance or Type Methods, Organizing Code using Extensions
Practical Animations - Uses of Core Animation in Apps, Animation Closures, The Transform Property, Animation at work using Music App Template.
HTTP and URL - Creating URLs, Creating and Executing a Network Request, Processing the Response, Work with API, Modify a URL with URL Components, Decoding JSON - JSON Basics, Decoding into Swift Types, Decoding into Custom Model Objects, Write a Completion Handler, Addressing Failure, Concurrency - Concurrency and Grand Central Dispatch, App Transport Security and HTTP Protocol

Recommended Resources / Links:

- <https://books.apple.com/us/book/develop-in-swift-data-collections/id1581183203>
- <https://www.youtube.com/@iOSAcademy>

Unit 4

Collection View - Anatomy of a Collection View, Collection View Layout, Emoji Dictionary App, Swift Generics - Generic Types, Generic Functions
Dynamic Data - Adding a Search Controller, Handling Data Changes, Diffable Data Source
Compositional Layout Components, Supplementary and Decoration Views, Multiple Layouts.

Recommended Resources / Links:

- <https://books.apple.com/us/book/develop-in-swift-data-collections/id1581183203>
- <https://www.youtube.com/@iOSAcademy>

Unit 5

Guided Projects - List (using Tables and Persistence), Restaurant (using Web), Habits (using Advanced Data Display), App Personality, App Icons Best Practices, Launch Screens, Color, Images and Icons, Animation, Typography, The Design Cycle

Recommended Resources / Links:

- <https://books.apple.com/us/book/develop-in-swift-data-collections/id1581183203>
- <https://www.youtube.com/@iOSAcademy>

Text Books:

1. Develop in Swift Data Collections, Apple Books

Course Outcomes:

At the end of the course, the student will be able to:

1. Create lists using scroll view and table view controllers (PO – 1, 2, 3, 5)(PSO – 1, 2)
2. Use system view controllers and build complex views (PO – 1, 2, 3, 4, 5)(PSO – 1, 2)
3. Encode and decode data to and from web (PO – 1, 2, 3, 5)(PSO – 1, 2)
4. Work with Collection Views (PO – 1, 2, 3, 5)(PSO – 1, 2)
5. Design, prototype and architect a project (PO – 1, 2, 3, 4, 5)(PSO – 1, 2)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
MCQ / Surprise Test	10	CO1, CO2, CO3, CO4, CO5
Programming Assignment	10	CO1, CO2, CO3, CO4, CO5
Semester-End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5

INTERNET OF THINGS	
Course Code: ISE632	Credits: 3:0:0
Pre – requisites: Micro Controller, CN	Contact Hours:42
Course Coordinator: Dr.Jagadeesh Sai D	

Course Contents

Unit 1

Introduction to Internet of Things Definition & Characteristics of IoT, Physical Design of IoT Things in IoT, IoT Protocols, Logical Design of IoT, IoT Functional Blocks, IoT Communication Models, IoT Communication APIs, IoT Enabling Technologies, Communication Protocols, Embedded Systems IoT Levels & Deployment Templates, IoT Levels

Pedagogy: Conceptual Learning through NPTEL video lectures and discussion-based sessions

Links: <https://nptel.ac.in/courses/106105166>

Unit 2

IoT and M2M: Introduction, M2M, Difference between IoT and M2M, SDN and NFV for IoT, Software Defined Networking, Network Function Virtualization, IoT System Management with NETCONF-YANG, Need for IoT Systems Management, Simple Network Management Protocol (SNMP)

Pedagogy: Case-based Learning using NPTEL videos and real-world IoT-M2M comparisons

Links: <https://nptel.ac.in/courses/106105166>

Unit 3

IoT Platforms Design Methodology: IoT Design Methodology, Purpose & Requirements Specification, Process Specification, Domain Model Specification, Information Model Specification, Service Specifications, IoT Level Specification, Functional View Specification, Operational View Specification, Device & Component Integration, Application Development, IoT Systems - Logical Design using Python, Functions Modules, Packages, File Handling Operations Classes, Python Packages of Interest for IoT.

Pedagogy: Hands-on Learning with NPTEL tutorials and guided Python-based implementation

Links: <https://nptel.ac.in/courses/106105224>

Unit 4

Raspberry Pi, About the Board, Linux on Raspberry Pi, Raspberry Pi Interfaces, Serial SPI, I2C, Programming Raspberry Pi with Python, Controlling LED with Raspberry Pi, interfacing an LED and Switch with Raspberry, Interfacing a Light Sensor (LDR) with Raspberry Pi, Other IoT Devices, pcDuino, Beagle Bone Black, Cubie board.

Pedagogy: Experiential Learning using Coursera videos and Raspberry Pi practical labs

Links: <https://www.coursera.org/learn/raspberry-pi-platform>

Unit 5

Python Web Application Framework – Django, Django Architecture, Starting Development with Django, designing a RESTful Web API, Amazon Web Services for IoT, Amazon EC2, Amazon Auto Scaling, Amazon S3, Amazon RDS Amazon Dynamo DB, Amazon Kinesis, Amazon SQS, Amazon EMR, SkyNet IoT Messaging Platform, INTEL Gen2, UDDO Board example

Pedagogy: Application-oriented Learning via YouTube tutorials and cloud-based integration projects

Links: <https://www.youtube.com/watch?v=F5mRW0jo-U4>

Text Books:

1. Internet of Things (A Hands-on-Approach) by Arshdeep Bagha, Vijay Madiseti University press, second edition 2019.

Reference Books:

1. IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things – David Hanes et al, Cisco Press, 2017, ISBN: 978-1-58714-456-1
2. Architecting the Internet of Things, Dieter Uckelmann, Mark Harrison, Florian Michahelles, Springer Berlin Heidelberg, 2021, ISBN: 978-3-642-19156-5

Self-Learning: Project Based Learning**Course Outcomes:**

1. Understand the design issues and fundamentals of IoT (PO-1,9,10,11) (PSO-2,3)
2. Design various methodologies for M2M and SDN architectures. (PO-1,9,10,11) (PSO-2,3)
3. Distinguish different cloud-based solution for IoT (PO-1,2,9,10,11) (PSO-2,3)
4. Develop IoT-based solutions for real-world problems. (PO-1,2,3,4,5,6,9,10,11) (PSO-2,3)
5. Analyze the various data analytical tools in IoT (PO-1,2,9,10,11) (PSO- 2,3)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2, CO3
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
MCQ	10	CO1, CO2, CO3, CO4, CO5
Project	10	CO1, CO2, CO3, CO4, CO5
Semester-End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5

BLOCK CHAIN ESSENTIALS AND DAPPS	
Course Code: ISE633	Credits: 3:0:0
Pre – requisites: NA	Contact Hours: 42
Course Coordinator: Mr. Prashanth Kambli	

Course Contents

Unit 1

Blockchain 101 : The growth of blockchain technology, Distributed systems, The history of blockchain and Bitcoin, Blockchain Consensus, CAP theorem and blockchain.

Recommended Resources / Links:

1. <https://nptel.ac.in/courses/106104220>
2. <https://blockgeeks.com/guides/what-is-blockchain-technology/>
3. Introduction to Blockchain Technology and Applications by Prof. Sandeep Shukla, IIT Kanpur:"

Unit 2

Decentralization: Decentralization using blockchain, Methods of decentralization, Routes to decentralization, Pertinent terminology, Platforms for decentralization: Ethereum, MaidSafe, Lisk, Innovative trends: Decentralized web.

Symmetric Cryptography: Introduction, Cryptographic primitives, Advanced Encryption Standard (AES)- How AES works

Public Key Cryptography: Mathematics, Asymmetric cryptography.

Recommended Resources / Links:

1. <https://nptel.ac.in/courses/106104220>
2. <https://blockgeeks.com/guides/what-is-blockchain-technology/>
3. Introduction to Blockchain Technology and Applications by Prof. Sandeep Shukla, IIT Kanpur:"

Unit 3

Public key Cryptography: Cryptographic constructs and blockchain technology

Introducing Bitcoin: Bitcoin- an overview, Cryptographic keys, Transactions, Blockchain, Mining

The Bitcoin Network and Payments: The Bitcoin network, Wallets

Recommended Resources / Links:

1. <https://nptel.ac.in/courses/106104220>
2. <https://blockgeeks.com/guides/what-is-blockchain-technology/>
3. Introduction to Blockchain Technology and Applications by Prof. Sandeep Shukla, IIT Kanpur:"

Unit 4

Smart contracts: History, Definition, Ricardian contracts, Smart contract templates, Oracles, Deploying smart contracts on blockchain, The DAO

Ethereum 101: Ethereum – an overview, The Ethereum network, Components of Ethereum ecosystem, The Ethereum Virtual Machine (EVM)

Recommended Resources / Links:

1. <https://nptel.ac.in/courses/106104220>
2. <https://blockgeeks.com/guides/what-is-blockchain-technology/>
3. Introduction to Blockchain Technology and Applications by Prof. Sandeep Shukla, IIT Kanpur:"

Unit 5

Further Ethereum: Blocks and blockchain, Wallets and client software, Nodes and miners

Hyperledger: Projects under Hyperledger, Hyperledger reference architecture, Hyperledger fabric, Hyperledger Sawtooth.

Recommended Resources / Links:

1. <https://nptel.ac.in/courses/106104220>
2. <https://blockgeeks.com/guides/what-is-blockchain-technology/>
3. Introduction to Blockchain Technology and Applications by Prof. Sandeep Shukla, IIT Kanpur:"

Text Books:

1. Imran Bashir, Mastering Blockchain, Third Edition, Packt, 2020

Reference Books:

1. Mastering Bitcoin: Programming the Open Blockchain Paperback – 2017 by Andreas M. O’rielly

Course Outcomes:

1. Illustrate the Blockchain terminologies with its applications. design (PO – 1) (PSO – 2)
2. Analyze the working principles of Blockchain (PO – 1, 2, 9, 10, 12) (PSO – 2)
3. Comprehend the principles and methodologies used in Bitcoin (PO – 1, 9, 10, 12) (PSO – 2)
4. Illustrate the Smart Contract Lifecycle (PO – 1, 2, 3, 9, 10, 12) (PSO – 2)
5. Create Ethereum Network, Wallets, Nodes, Smart contract & DApps also develop Blockchain Based Application Architecture using Hyperledger (PO – 1, 2, 3, 5, 9, 10, 12) (PSO – 2)

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2, CO3
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
Quiz	10	CO1, CO2, CO3, CO4,CO5
Tool Demonstration	10	CO1, CO2, CO3, CO4,CO5
Semester-End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4,CO5

SYSTEM SIMULATION AND MODELING	
Course Code: ISE634	Credits: 3:0:0
Pre – requisites: Engineering Mathematics	Contact Hours: 42
Course Coordinator: Mushtaq Ahmed D M	

Course Contents

Unit 1

Introduction to Simulation: Simulation, Advantages and disadvantages, Areas of application, System environment, Components of a system, Discrete and continuous systems, Model of a system, Types of Models, Steps in a Simulation Study.

Simulation examples: Simulation of queuing systems, Simulation of inventory systems. Other simulation examples.

Recommended Resources / Links:

1. <https://archive.nptel.ac.in/courses/112/107/112107220/>
2. <https://a.impartus.com/ilc/#/course/1259594/1112>

Unit 2

General Principles, Simulation Software: Concepts in Discrete-Event Simulation: The Event-Scheduling / Time-Advance Algorithm, World Views, Manual simulation Using Event Scheduling; List processing. Simulation in Java; Simulation in GPSS.

Statistical Models in Simulation: Review of terminology and concepts; Useful statistical models, Discrete Distributions, Continuous Distributions, Poisson Process.

Recommended Resources / Links:

1. <https://archive.nptel.ac.in/courses/112/107/112107220/>
2. <https://a.impartus.com/ilc/#/course/1259594/1112>

Unit 3

Queuing Models: Characteristics of queuing systems, queuing notation, Long-run measures of performance of queuing systems, Steady-state behavior of M/G/1 queue, Multiserver Queue: $M|M|C|\infty|\infty$, Networks of queues.

Random-Number Generation: Properties of random numbers, Generation of pseudo-random numbers; Techniques for generating random numbers; Tests for Random Numbers-Frequency test, Run test, Autocorrelation test.

Recommended Resources / Links:

1. <https://archive.nptel.ac.in/courses/112/107/112107220/>
2. <https://a.impartus.com/ilc/#/course/1259594/1112>

Unit 4

Random Variate Generation: Inverse transform Technique-Exponential, Uniform, Weibull, and Triangular Distribution, **Acceptance-Rejection technique:** Poisson Distribution, Convolution method.

Input Modelling: Data Collection, Identifying the distribution with data, Parameter estimation, Goodness of Fit Tests, selecting input models without data, Multivariate and Time Series analysis.

Recommended Resources / Links:

1. <https://archive.nptel.ac.in/courses/112/107/112107220/>

2. <https://a.impartus.com/ilc/#/course/1259594/1112>

Unit 5

Verification and Validation of Simulation Models: Model building, verification and validation, Verification of simulation models, Calibration and validation of models.

Output Analysis: Types of simulations with respect to output analysis; Stochastic nature of output data, Absolute measures of performance and their estimation, Output analysis for terminating simulations, Output analysis for steady-state simulations.

Recommended Resources / Links:

1. <https://archive.nptel.ac.in/courses/112/107/112107220/>
2. <https://a.impartus.com/ilc/#/course/1259594/1112>

Text Books:

1. Jerry Banks, John S. Carson II, Barry L. Nelson, David M. Nicol: Discrete-Event System Simulation, Fifth Edition, Pearson Education 2013.

Reference Books:

1. Lawrence M. Leemis, Stephen K. Park: Discrete – Event Simulation: A First Course, Pearson Prentice-Hall 2015.
2. Sheldon M. Ross: Simulation, Sixth Edition, Elsevier 2022.
3. Averill M. Law: Simulation Modelling and Analysis, Sixth Edition, Tata McGraw-Hill 2024.

Course Outcomes:

1. Explain modeling and simulation types, entities, objectives and benefits. (PO-1, 2) (PSO-1, 2)
2. Recognize the simulation types and steps for variety of complex systems. (PO-1, 2, 3) (PSO-1, 2)
3. Construct simulation models from enterprise high level models. (PO-1,2, 3, 4, 5) (PSO-1, 2)
4. Execute efficient simulation to analyze real-world systems. (PO-1,2, 3, 5) (PSO-1, 2)
5. Evaluate the results of simulation and analysis to improve or optimize systems. (PO-2, 3) (PSO-2)

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2, CO3
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
Quiz/MOOC	10	CO1, CO2, CO3
Case Studies	10	CO4, CO5
Semester-End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4 , CO5

BUSINESS ANALYTICS	
Course Code: ISE641	Credits: 3:0:0
Pre – requisites: NIL	Contact Hours:42
Course Coordinator: Dr Pushpalatha M N	

Course Contents

Unit 1

Introduction: Why Analytics, Business Analytics: The Science of data driven decision making, Descriptive Analysis, Predictive Analytics, Prescriptive Analytics, Big Data Analytics, Web and Social Media Analytics, Machine Learning Algorithms, Framework for data driven decision making, Analytics Capability Building, Roadmap, Challenges, Types (Descriptive, Predictive and Prescriptive).

Recommended Resources / Links:

1. <https://www.youtube.com/watch?v=cWWc97wxS20>
2. https://www.youtube.com/watch?v=bajBth_C4w4

Unit 2

Descriptive Analytics: Data Types and Scales, Types of Data Measurement Scales, Population and Sample, Measures of Central Tendency, Percentile, Decile, and Quartile, Measures of Variation, Measures of Shape –Skewness and Kurtosis, Data Visualization.

Recommended Resources / Links:

1. <https://www.youtube.com/watch?v=pFZP8nws5jI>
2. <https://www.youtube.com/watch?v=1MiT06JFNo4>
3. <https://www.youtube.com/watch?v=6lQn1hdG43o&t=1047s>

Unit 3

Data Mining: Data Mining, application of data mining, Anomaly Detection, Association Rule Learning, Cluster Analysis, Statistical Classification, Regression Analysis, Automatic Summarization, Examples of Data Mining.

Recommended Resources / Links:

1. <https://www.youtube.com/watch?v=81bm2OsEzbg>
2. <https://www.youtube.com/watch?v=ykZ-UGcYWg&list=PLLspfyOYoQcI6Nno3gPq0h5YSe81hsc>
3. https://www.youtube.com/watch?v=i5Oyzn3XryY&list=PLm5IZukmvRNmj2eqq9Y0_7GRP_4dozIGZ
4. <https://www.youtube.com/watch?v=XO97Uon83Os>

Unit 4

Data Warehousing: Data Warehouse, Data Mart, Master Data Management, Dimension (Data Warehouse), Slowly Changing Dimension, Data Vault Modeling, Extract, Transform, Load, Star Schema, Mapping problems to machine learning tasks, Evaluating models, Validating models.

Recommended Resources / Links:

1. <https://www.youtube.com/watch?v=m-aKj5ovDfg>
2. https://www.youtube.com/watch?v=oF_2uDb7DvQ

Unit 5

Essential Aspects of Business Intelligence: Context Analysis, Business Performance Management, Business Process Discovery, Information System, Organizational Intelligence, Data Visualization,

Data Profiling, Data Cleansing, Process Mining, Competitive Intelligence, Operational Intelligence, Business Activity Monitoring, Complex Event Processing, Business Process Management, Metadata, Root Cause Analysis.

Recommended Resources / Links:

1. <https://www.youtube.com/watch?v=dRG5JP6zxck>
2. https://www.youtube.com/watch?v=EeUeuz_Exe0
3. <https://www.youtube.com/watch?v=yJE90Mj7JFY>

Text Books:

1. Drew Bentley, “Business Intelligence and Analytics” Library Press publication -2017
2. U. Dinesh kumar, “Business Analytical – The science of data driven decision making”, Wiley 2017

Reference Books:

1. Sharda R, Delen D, Turban E, Aronson J, Liang T. P, (2014), Business Intelligence and Analytics: Systems for Decision Support, 10th edition, Pearson Education.
2. Glenn J. Myatt, “Making Sense of Data : A Practical Guide to Exploratory Data Analysis and Data Mining”, John Wiley & Sons, Second Edition, 2014.
3. Carlo-Vercellis, “Business intelligence datamining and optimization for decision making”, First Edition.
4. An Introduction to Business Analytics, Ger Koole, Lulu.com, 2019.

Course Outcomes:

1. Understand the basic concepts of Business Analytics, Types, Framework and Challenges. (PO-1,2,12)(PSO-3)
2. Understand the emergence of descriptive analytics as a competitive strategy.(PO-1,2,3,9,10,12)(PSO-3)
3. Apply data mining functionalities on real world problems and Explain the analyzing techniques of various data.(PO-1,2,3,12)
4. Examine data warehousing concepts, Star and Snowflake Schemas, ETL processes, and manage Slowly Changing Dimensions (SCDs).(PO-1,2,3,12)(PSO-3)
5. Understand various Essential Aspects of Business Intelligence.(PO-1,2,9,10,12)(PSO-3)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2, CO3
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
Course Project	10	CO1, CO2, CO3, CO4, CO5
Quiz	10	CO1, CO2, CO3, CO4
Semester-End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5

DATA ENGINEERING AND MLOPS	
Course Code: ISE642	Credits: 3:0:0
Pre - requisites: Nil	Contact Hours: 42
Course Coordinator: Dr. Sumana M	

Course Content

Unit I

Data Engineering: Definition, The Data Engineering Lifecycle, Evolution of the Data Engineer, Data Engineering and Data Science, Data Engineering Skills and Activities, Data Maturity and the Data Engineer, The Background and Skills of a Data Engineer, Business Responsibilities, Technical Responsibilities, The Continuum of Data Engineering Roles, Data Engineers Inside an Organization, Internal-Facing Versus External-Facing Data Engineers, Data Engineers and Other Technical Roles, Data Engineers and Business Leadership.

Data Engineering Lifecycle: The Data Lifecycle Versus the Data Engineering Lifecycle, Generation: Source Systems, Major Undercurrents Across the Data Engineering Lifecycle

Unit II

Data Architecture: Enterprise Architecture Defined, Data Architecture Defined, “Good” Data Architecture, Principles of Good Data Architecture, Major Architecture Concepts, Domains and Services, Distributed Systems, Scalability, and Designing for Failure, Tight Versus Loose Coupling: Tiers, Monoliths, and Microservices, User Access: Single Versus Multitenant, Event-Driven Architecture, Examples and Types of Data Architecture

Choosing Technologies Across the Data Engineering Lifecycle: Team Size and Capabilities, Speed to Market, Interoperability, Cost Optimization and Business Value, Total Cost of Ownership Total Opportunity Cost of Ownership, FinOps, Today Versus the Future: Immutable Versus Transitory Technologies: Hybrid Cloud, Multicloud , Decentralized: Blockchain and the Edge ,Monolith Versus Modular , Serverless Versus Servers, Server Versus Serverless evaluation

Unit III

MLOps Challenges, MLOps to Mitigate Risk, Risk Assessment, Risk Mitigation, MLOps for Responsible AI, MLOps for Scale.

Key MLOps Features: Model Development, Establishing Business Objectives, Data Sources and Exploratory Data Analysis, Feature Engineering and Selection, Training and Evaluation, Reproducibility, Responsible AI, Productionalization and Deployment, Model Deployment Types and Contents, Model Deployment Requirements, Monitoring

Developing Models: Machine Learning Model, Required Components, Different ML Algorithms, Different MLOps Challenges, Data Exploration, Feature Engineering and Selection, Feature Engineering Techniques, How Feature Selection Impacts MLOps Strategy, Experimentation, Evaluating and Comparing Models, Choosing Evaluation Metrics, Cross-Checking Model Behavior, Impact of Responsible AI on Modeling, Version Management and Reproducibility

Unit IV

Preparing for Production: Runtime Environments, Adaptation from Development to Production

Environments, Data Access Before Validation and Launch to Production, Final Thoughts on Runtime Environments, Model Risk Evaluation, The Purpose of Model Validation, The Origins of ML Model Risk, Quality Assurance for Machine Learning

Deploying to Production: CI/CD Pipelines, Building ML Artifacts, The Testing Pipeline, Deployment Strategies, Categories of Model Deployment, Considerations When Sending Models to Production, Maintenance in Production, Containerization, Scaling Deployments, Requirements and Challenges

Unit V

Monitoring and Feedback Loop: Models Be Retrained, Understanding Model Degradation, Ground Truth Evaluation, Input Drift Detection, Drift Detection in Practice, Example Causes of Data Drift, Input Drift Detection Techniques, The Feedback Loop, Logging, Model Evaluation, Online Evaluation

Model Governance: Governance the Organization Needs, Matching Governance with Risk Level, Current Regulations Driving MLOps Governance, Pharmaceutical Regulation in the US: GxP Financial Model Risk Management Regulation, GDPR and CCPA Data Privacy Regulations, The New Wave of AI-Specific Regulation, The Emergence of Responsible AI, Key Elements of Responsible AI (Element 1 to element 5), A Template for MLOps Governance (Step 1 to 8)

Text Books:

1. Joe Reis, Matt Housley Fundamentals of Data Engineering: Plan and Build Robust Data Systems, O'Reilly, 2022
2. Mark Treveil and the Dataiku Team. Introducing MLOps How to Scale Machine Learning in the Enterprise, O'Reilly, 2020

Course Outcomes (COs):

At the end of the course, the student will be able to:

1. Understand data engineering life cycle (PO – 1, 4, PSO - 3)
2. Choose technologies across data engineering life cycle (PO – 2, 4, PSO - 3)
3. Evaluate machine learning models for MLOps (PO – 2, 4, PSO - 3)
4. Understand MLOps deployment strategies (PO – 1, 2, PSO - 3)
5. Understand MLOps model governance (PO – 1, 2, PSO - 3)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
Programming Assignment	10	CO1, CO2, CO3, CO4, CO5
MCQ Test	10	CO1, CO2, CO3, CO4, CO5
Semester-End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5

NATURAL LANGUAGE PROCESSING	
Course Code: ISE643	Credits: 3:0:0
Pre – requisites: NIL	Contact Hours:42L
Course Coordinator: Dr. Rajeshwari S B	

Course Contents

Unit 1

Introduction: Knowledge in Speech and Language Processing, Ambiguity, Models and Algorithms; Language, Thought, and Understanding; The State of the Art and The Near-Term Future; **Regular Expressions and Automata; Morphology and Finite-State Transducers:** Inflectional and Derivational Morphology, Lexicon-free FSTs: The Porter Stemmer, Human Morphological Processing.

Recommended Resources / Links:

1. <https://nptel.ac.in/courses/109106083>
2. <https://nptel.ac.in/courses/106101007>
3. <https://media.dev.nptel.ac.in/content/mp4/106/106/106106211/MP4/mod01lec01.mp4>

Unit 2

N-grams: Counting Words in Corpora, Smoothing, N-grams for Spelling and Pronunciation, Entropy;

Word Classes and Part-of-Speech Tagging: Part-of-Speech Tagging, Rule-based Part-of-speech Tagging, Stochastic Part-of-speech Tagging, Transformation-Based Tagging.

Recommended Resources / Links:

1. <https://nptel.ac.in/courses/106101007>
2. <https://media.dev.nptel.ac.in/content/mp4/106/106/106106211/MP4/mod01lec02.mp4>
3. <https://media.dev.nptel.ac.in/content/mp4/108/105/108105572/MP4/mod01lec04.mp4>

Unit 3

Context-Free Grammars for English: Constituency, Context-Free Rules and Trees, Sentence-Level Constructions. **Parsing with Context-Free Grammars:** The Earley Algorithm; **Features and Unification:** Feature Structures, Unification of Feature Structures, Features Structures in the Grammar, Implementing Unification, Parsing with Unification Constraints.

Recommended Resources / Links:

1. <https://media.dev.nptel.ac.in/content/mp4/108/105/108105572/MP4/mod01lec03.mp4>
2. <https://nptel.ac.in/courses/106105158>
3. <https://nptel.ac.in/courses/109106083>

Unit 4

Lexicalized and Probabilistic Parsing: Probabilistic Context-Free Grammars, Problems with PCFGs. **Representing Meaning:** First Order Predicate Calculus, Some Linguistically Relevant Concepts, Related Representational Approaches, Alternative Approaches to Meaning.

Recommended Resources / Links:

1. <https://nptel.ac.in/courses/106105158>
2. <https://nptel.ac.in/courses/106101007>
3. <https://media.dev.nptel.ac.in/content/mp4/106/106/106106211/MP4/mod01lec03.mp4>

Unit 5

Semantic Analysis: Syntax-Driven Semantic Analysis. **Lexical Semantics:** Relations Among Lexemes and Their Senses, WordNet: A Database of Lexical Relations. **Discourse:** Reference Resolution, Text Coherence, Discourse Structure.

Recommended Resources / Links:

1. <https://nptel.ac.in/courses/106101007>

2. <https://nptel.ac.in/courses/106105158>
3. <https://nptel.ac.in/courses/106106211>

Text Book:

1. Daniel Jurafsky and James H Martin, “Speech and Language Processing: An introduction to Natural Language Processing, Computational Linguistics and Speech Recognition”, Prentice Hall, 2nd Edition, 2013.

Reference Books:

1. Tanveer Siddiqui, U.S. Tiwary, “Natural Language Processing and Information Retrieval”, Oxford University Press, 2008.
2. Steven Bird, Ewan Klein, Edward Loper, “Natural Language Processing with Python- Analyzing Text with the Natural Language Toolkit”, O’reilly Publications , 2009.

Course Outcomes:

1. Illustrate how formal models are used in speech and language technologies to represent and process linguistic knowledge. (PO- 1, 2, 3) (PSO-1, 3)
2. Describe the role of N-gram models and POS tagging in performing spell checking and handling pronunciation-related tasks. (PO- 2, 3, 5, 9, 10) (PSO-1, 3)
3. Depict how unification and early parsing algorithms help in capturing the syntactic structure of a text. (PO- 2, 3, 5, 9, 10) (PSO-1, 3)
4. Emphasize on how probabilistic context free grammar and first order predicate calculus are used to represent meaning in language processing.(PO- 1, 2, 3, 5, 9, 10) (PSO-1, 3)
5. Apply syntax-driven methods to perform semantic analysis, explore lexical semantics using WordNet, and analyze discourse through reference resolution & coherent text understanding. (PO- 2, 3, 5, 9, 10) (PSO-1, 3)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
MiniProject on NLP using Quantum Computing	10	CO1, CO2, CO3, CO4, CO5
MCQ Test	10	CO1, CO2, CO3, CO4, CO5
Semester-End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5

CLOUD COMPUTING	
Course Code: ISE644	Credits: 3:0:0
Pre – requisites: Nil	Contact Hours: 42
Course Coordinator: Dr. S R Mani Sekhar	

Course Contents

Unit 1

Introduction: Network centric computing and network centric content, Peer-to-peer systems, Cloud Computing, Cloud Computing delivery models & Services, Ethical issues, Cloud vulnerabilities, Challenges. **Cloud Infrastructure:** Amazon, Google, Azure & online services, open source private clouds. Service level and compliance level agreement, Responsibility sharing

Recommended Resources / Links:

- https://onlinecourses.nptel.ac.in/noc21_cs14/preview
- <https://www.youtube.com/watch?v=36zducUX16w>
- <https://www.youtube.com/watch?v=V2DCYO-sWRQ>

Unit 2

Cloud Computing: Applications & Paradigms, Challenges, existing and new application opportunities, Architectural styles of cloud applications, Workflows: Coordination of multiple activities, Coordination based on a state machine model – the ZooKeeper, High performance computing on a cloud, cloud computing for biological research

Recommended Resources / Links:

- <https://www.youtube.com/watch?v=QWtKykvDvQ>
- <https://www.youtube.com/watch?v=FdIdX13BF7Q>
- https://onlinecourses.nptel.ac.in/noc21_cs14/preview
- <https://www.youtube.com/watch?v=AS5a91DOmks>

Unit 3

Cloud Resource Virtualization: Virtualization, Layering and virtualization, Virtual machine monitors, Virtual machines, Performance and security isolation, Full virtualization and paravirtualization, Case study: *Xen* -a VMM based on paravirtualization, A performance comparison of virtual machines, The darker side of virtualization.

Recommended Resources / Links:

- https://onlinecourses.nptel.ac.in/noc21_cs14/preview
- <https://www.youtube.com/watch?v=Sb5SO3WRSws>
- <https://www.youtube.com/watch?v=WIPi7l0d8Ww>

Unit 4

Cloud Resource Management and Scheduling: Policies and mechanisms for resource management, Applications of control theory to task scheduling on a cloud, Stability of a two-level resource allocation architecture, **Resource bundling:** combinatorial auctions for cloud resources, Scheduling algorithms for computing clouds, fair queuing.

Recommended Resources / Links:

- https://onlinecourses.nptel.ac.in/noc21_cs14/preview
- https://www.youtube.com/watch?v=MzclA_hdNLY
- https://www.youtube.com/watch?v=BHNbbqE_ttc

Unit 5

Storage systems: Storage models, file systems, databases, DFS, GFS, Locks & Chubby. **Cloud security:** Risks, Security, privacy and privacy impacts assessments, Trust, VM Security,

Security of virtualization.

Recommended Resources / Links:

- https://onlinecourses.nptel.ac.in/noc21_cs14/preview
- <https://www.youtube.com/watch?v=O-XBhVv2pgE>
- <https://www.youtube.com/watch?v=sF014G2BUXM>

Text Books:

1. Dan Marinescu, Cloud Computing: Theory and Practice, 2nd edition, MK Publishers 2018

Reference Books:

1. Kai Hwang, Jack Dongarra, Geoffrey Fox, Distributed and Cloud Computing, From Parallel Processing to the Internet of Things, 1st edition, MK Publishers, 2012.
2. Anthony T. Velte, Toby J. Velete, Robert Elsenpeter, Cloud Computing: A Practical Approach, Tata McGraw Hill.

Course Outcomes:

At the end of the course, the student will be able to:

1. Apply the concepts of cloud delivery models and services. (PO-1,2,3,5,7,9,10,12) (PSO-2,3)
2. Build various cloud based applications. (PO-1,2,3,5,7,9,10,12) (PSO-2,3)
3. Illustrate different cloud resource virtualization strategies with case studies. (PO-1,7) (PSO- 2,3)
4. Describe cloud resource management and scheduling policies (PO-1 ,7) (PSO-2,3)
5. Create cloud instances by applying storage models and security aspects. (PO- 1,2,3,5,7,9,10,12) (PSO-2,3)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
Tool / Assignment	20	CO1, CO2, CO3, CO4, CO5
Semester-End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5

DATA ANALYTICS WITH PYTHON	
Course code: ISOE14	Credits: 3:0:0
Prerequisites: NIL	Contact Hours : 42
Course coordinator(s): Mrs. Kavya K S	

Course Contents:

Unit – 1

Introduction to Python Programming: Introduction to Python programming language: syntax, variables, data types, Control structures: loops and conditional statements, Functions and modules in Python

Introduction to Python libraries for data analytics: NumPy: Arrays, mathematical operations, array manipulation Pandas: Data structures (Series, DataFrame), data manipulation, data alignment, indexing, grouping, merging

Pedagogy/Course delivery tools:-Chalk and talk, Power Point Presentation, Videos, Jupyter notebook

Links:
<https://www.youtube.com/watch?v=eWRfhZUzrAc&t=1s>
<https://www.youtube.com/watch?v=JXY7KGOZ89U>
<https://youtu.be/2uvysYbKdjM>

Unit – 2

Data Loading, Storage, and File Formats: Reading and writing data from various sources: CSV, Excel, JSON, SQL databases

Data cleaning techniques: Importing and exporting data in Python, Handling missing values: imputation, deletion, Handling outliers: detection, removal, transformation, Data normalization and scaling

Data transformation techniques: Encoding categorical variables, Feature scaling and normalization, Feature extraction and engineering

Pedagogy/Course delivery tools:-Chalk and talk, Power Point Presentation, Videos, Jupyter notebook

Links:
<https://www.youtube.com/watch?v=qkrGC7971fk>
<https://www.youtube.com/watch?v=cWf08xuSqdU>

Unit - 3

Data visualization: Summarizing and describing data: mean, median, mode, variance, standard deviation, Data visualization techniques: Bar plots, pie charts, Box plots, histogram, Scatter, violin plots, Heatmaps, pair plots

Data Aggregation: Column-wise and Multiple Function Application, Group-wise Operations and Transformations, Random Sampling and Permutation, Group Weighted Average and Correlation

Pedagogy/Course delivery tools:-Chalk and talk, Power Point Presentation, Videos, Jupyter notebook

Links:
<https://www.youtube.com/watch?v=TXjWrUCErAE>
https://www.youtube.com/watch?v=W_YUk8IvsqI
<https://www.youtube.com/watch?v=FSHyTSnKDjU>

Unit – 4

Time series: Date and Time Data, Types and Tools, Converting between string and datetime, Date Ranges, Frequencies and Shifting, Down sampling, Up sampling and Interpolation, Time Series Plotting, Moving Window Functions

Pedagogy/Course delivery tools:-Chalk and talk, Power Point Presentation, Videos, Jupyter notebook Links:

<https://www.youtube.com/watch?v=eKiXtGzEjos>

<https://www.youtube.com/watch?v=UFuo7EHI8zc>

Unit – 5

Machine Learning with Python: Supervised learning vs. unsupervised learning, Classification vs. regression, **Supervised learning algorithms:** Linear regression, Logistic regression, Decision trees, Random forests, Support Vector Machines (SVM)

Unsupervised learning algorithms: K-means clustering, Hierarchical clustering, Principal Component Analysis (PCA)

Model evaluation and validation techniques: Train-test split, Cross-validation, Performance metrics: accuracy, precision, recall, F1-score, ROC curve, AUC

Pedagogy/Course delivery tools:-Chalk and talk, Power Point Presentation, Videos, Jupyter notebook Links:

https://www.youtube.com/watch?v=wvODQqb3D_8

<https://www.youtube.com/watch?v=DDGjz6lsI-k>

<https://www.youtube.com/watch?v=hDKCxebp88A>

Text Books:

1. Learning Python, Fourth Edition, Mark Lutz, O'Reilly publication
2. Author: Wes McKinney, "Python for Data Analysis", ISBN-13: 978-1491957660, 2nd Edition by O'Reilly Media
3. Introduction to Machine Learning with Python, A guide for Data Scientists, Andreas C. Müller & Sarah Guido, O'Reilly Publications

Reference:

1. Doing Data Science Cathy O'Neil and Rachel Schutt Straight Talk from The Frontline. O'Reilly 2014

Course outcomes:

The students will be able to

- Identify Data analytics problem for different domains (PO 2) (PSO 3)
- Describe the basic programming concepts of python and Functions (PO 1) (PSO 3)
- Apply appropriate data types/ data structures for the given problem using Lists, Dictionaries, Tuples. (PO 1, 2, 3,5,9,10,12) (PSO 3)
- Apply python to solve different data analytics problems and visualize the data. (PO 1, 2, 3,5,9,10,12) (PSO 3)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2, CO3
Internal Test-II	30	CO3, CO4, CO5

Average of the two internal tests shall be taken for 30 marks.		
Other Components		
Project implementation	10	CO1, CO2, CO3, CO4, CO5
Project presentation and Report	10	CO1, CO2, CO3, CO4, CO5
Semester-End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5

Fundamentals of Database Management Systems	
Course Code: ISOE15	Credits: 3:0:0
Pre – requisites: NIL	Contact Hours: 42
Course Coordinator: Lincy Mathews	

Course Contents

Unit 1

Introduction; An example; Characteristics of Database approach; Actors on the scene, Workers Behind the Scene, Advantages of using DBMS approach, Data models, Schemas and Instances; Three-schema architecture and data independence; Database languages and interfaces; Introduction to NoSQL, Types of NoSQL Database, NoSQL vs SQL Comparison.

Pedagogy: PPT, Chalk and Talk Method

Links:

- <https://nptel.ac.in/courses/106106220>
- <https://www.youtube.com/watch?v=OWX4RvijwLw&list=PLIwC9bZ0rmjSkm1VRJROX4vP2YMIf4Ebh&index=3>
- <https://www.youtube.com/watch?v=rrG7azSlyWI&list=PLIwC9bZ0rmjSkm1VRJROX4vP2YMIf4Ebh&index=4>
- <https://www.youtube.com/watch?v=DRSog3SA4Y&list=PLIwC9bZ0rmjSkm1VRJROX4vP2YMIf4Ebh&index=5>

Unit 2

Using High-Level Conceptual Data Models for Database Design; An Example Database Application; Entity Types, Entity Sets, Attributes, and Keys; Relationship types, Relationship Sets, Roles, and Structural Constraints; Weak Entity Types; Refining the ER Design; Case study on ER Diagrams (Example –Company Database).

Pedagogy: PPT, Chalk and Talk Method

Links:

- <https://www.youtube.com/watch?v=cx7rBTS2dBM&list=PLIwC9bZ0rmjSkm1VRJROX4vP2YMIf4Ebh&index=12>
- https://www.youtube.com/watch?v=pm_Tr3eZAac&list=PLIwC9bZ0rmjSkm1VRJROX4vP2YMIf4Ebh&index=13
- https://www.youtube.com/watch?v=pm_Tr3eZAac&list=PLIwC9bZ0rmjSkm1VRJROX4vP2YMIf4Ebh&index=14

Unit 3

Relational Model Concepts; Relational Model Constraints and Relational Database Schemas; Update Operations, Transactions, and dealing with constraint violations; Relational Database, Design using ER-to-relational Mapping, Case Study on ER-to-relational mapping. SQL Data Definition and Data Types; Specifying basic constraints in SQL; Schema change statements in SQL; Basic queries in SQL: where, aggregate functions, order by.

Pedagogy:

- Links:
<https://www.youtube.com/watch?v=w1XdPholzWY&list=PLIwC9bZ0rmjSkm1VRJROX4vP2YMIf4Ebh&index=6>
- <https://www.youtube.com/watch?v=ywUn5qNmxNo&list=PLIwC9bZ0rmjSkm1VRJROX4vP2YMIf4Ebh&index=7>

Unit 4

More complex SQL Queries. Insert, Delete, and Update statements in SQL; JOINS, Views (Virtual Tables) in SQL. Triggers, procedures, functions, MongoDB: Create collections, Inserting, Update, and Deleting the documents, basic query functions.

Pedagogy: PPT, Chalk and Talk Method

Links:

- <https://www.youtube.com/watch?v=ywUn5qNmxNo&list=PLIwC9bZ0rmjSkm1VRJROX4vP2YMIf4Ebh&index=8>

Unit 5

Informal Design Guidelines for Relation Schemas; Functional Dependencies; Normal Forms Based on Primary Keys; General Definitions of Second and Third Normal Forms, BCNF; Properties of Relational Decompositions;

Pedagogy: PPT, Chalk and Talk Method

Links:

- <https://www.youtube.com/watch?v=2tPNtnCfGzo&list=PLyqSpQzTE6M-xymXgqewlzcC3U4cdRoSu&index=24>
- <https://www.youtube.com/watch?v=Q5jbdZtnOjg&list=PLyqSpQzTE6M-xymXgqewlzcC3U4cdRoSu&index=22>

Text Books:

1. RamezElmasri and ShamkantB.Navathe: Fundamentals of Database Systems, 7th Edition, Addison-Wesley, 2016.
2. Benjamin Rosenzweig, Elena Silvestrova Rakhimov : Oracle PL/SQL by Example, 6th Edition, 2023.
3. https://www.tutorialspoint.com/mongodb/mongodb_tutorial.pdf

Reference Books:

1. Silberschatz, Korth, and Sudharshan: “Database System Concepts, Seventh Edition, Mc-GrawHill, 2019”.
2. Raghu Ramakrishnan, Johannes Gehrke, “Database Management Systems, Third Edition”, McGraw-Hill, 2003.
3. Kyle Banker Peter Bakkum Shaun Verch Douglas Garrett Tim Hawkins “MongoDB In Action 2nd Edition” , 2016

Self-Learning (Programming Assignment/ Case Study/Project-Based Learning/MOOC-based learning / GATE-oriented aptitude test / Simulation):

Class assignment is provided twice within the semester.

Course Outcomes:

CO1: Understand the fundamental concepts of databases, the DBMS approach, data models, and the differences between SQL and NoSQL databases. PO [1, 2, 3, 4, 5, 10, 12] PSO [1, 2]

CO2: Develop the ability to design databases using the Entity-Relationship (ER) model, including defining entities, relationships, and attributes, and translating ER diagrams into relational schemas. PO [1, 2, 3, 4, 5, 9, 10, 11, 12] PSO [1, 2, 3]

CO3: Gain proficiency in using SQL for defining, manipulating, and querying relational databases, as well as understanding relational database constraints and schema design. PO [1, 2, 3, 4, 5, 9, 10, 11, 12] PSO [1, 2, 3]

CO4: Master advanced SQL techniques, including joins, subqueries, and views, and acquire practical knowledge of MongoDB, including CRUD operations and database management. PO [1, 2, 3, 4, 5, 9, 10, 11, 12] PSO [1, 2, 3]

CO5: Understand and apply database normalization techniques, including functional dependencies and normal forms (1NF, 2NF, 3NF, BCNF), to design efficient and scalable relational schemas. PO [1, 2, 3, 4, 5, 9, 10, 11, 12] PSO [1, 2, 3]

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test - I	30	CO1, CO2, CO3
Internal Test - II	30	CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
Class Assignment 1	10	CO1, CO2, CO3
Class Assignment 2	10	CO4, CO5
Semester End Examination	100(Scale down to 50)	CO1, CO2, CO3, CO4.CO5

MINI-PROJECT	
Course Code: ISP67	Credits: 0:0:4
Pre – requisites: Nil	Contact Hours:
Course Coordinator: Internal Guide	

Course Contents

Guidelines:

Students have to work in a group of 3/4 to solve a problem in the specific domain. An Internal Guide is allotted per batch based on their domain of expertise who guides and monitors the project progress. The Internal Guide can arrange for doubt clarification classes if requested by his/her project student and records the same.

Following are the Rubrics considered for Evaluation of mini-project:

Relevance of Project: Student are expected to clearly state the relevance of project to current IT environment and Society in general

Literature Survey: Student need to study research articles/ existing projects to identify the gaps in the identified problem statement.

Design: Student should prepare design document by considering class/use case/component diagram/state model/sequence model/activity/Interaction model.

Implementation: Student need to implement the designed model using the suitable techniques.

Presentation: Periodically student need to present their progress in front of the evaluation committee. Depending upon the quality of ppt, depth of coverage, answering capabilities to questions raised and division of labor identified during presentation and team work, evaluation committee will be deciding score in this criteria for individual students.

Report: Each group need to prepare the project report and submit the same to the department. Reports need to be adhered to the standard format defined by the department

Phases of Evaluation for CIE:

Phases	Particulars	Marks(50)
Phase-I	Acceptance of Problem Statement, Objectives and Expected Outcome.	10
Phase-II	Literature Review, Requirement Specifications, Design & Methodology, Partial Implementation of the project	20
Phase-III	Implementation, Testing, Conclusion & Future Enhancement, References, Demonstration of the Project.	20

Course Outcomes (COs):

At the end of the course Students will be able to:

1. Identify a problem, review research literature and analyze requirements (POs – 1,12) (PSO - 1, 2, 3)

2. Schedule milestone and deliverables using appropriate project management techniques (POs – 8,9,10,11). (PSO - 1, 2, 3)
3. Design and implement the solution to selected problem using standard models and processes (POs – 1,2,3,4,5,6,7,8,9, 10,11,12) (PSO – 1,2, 3)
4. Analyze the results and produce substantial written documentation (POs – 1,2, 4,8,9, 10,11,12) (PSO – 1,2, 3)

RUBRICS FOR MINI PROJECT WORK

Maximum Marks: 50

Sl No	Performance indicators	Excellent	Good	Average	Poor
1.	Identification of Problem Domain and Motivation (5)	(1). Detailed and extensive explanation of the purpose and need of the project. (2). Detailed and extensive explanation of the specifications and the limitations of the existing system. (5 Marks)	Good explanation Of the purpose and need of the project. Collects a great deal of information and a good study of the existing systems. (3 Marks)	Average Explanation about the purpose and need of the project. A moderate study of the existing systems collects some basic information.(2 Marks)	Moderate explanation of the purpose and need of the project. The explanation of the specifications and the limitations of the existing systems is not very satisfactory, limited information. (1 Mark)
2.	Novelty of Project(4)	Demonstration of novel or unique and innovative ideas, designs, products, or approaches to generate new knowledge or creative works. The student has created their own design or product with originality.(4 marks)	Application of the existing ideas in the literature with some originality. • Attempts to create a novel or unique idea, design, product, or approach. • Reformulates an idea or collection of available ideas or approaches with very little originality shown. (2 marks)	Application of the existing ideas in the literature with some originality. • Attempts to create a novel or unique idea, design, product, or approach. • Reformulates an idea or collection of available ideas or approaches with very little originality shown.(1 mark)	Creativity and originality is very limited.
2	Detailed analysis of Objectives (5)	All objectives of the proposed work are well defined. (5 Marks)	A good justification for the objectives. (3 Marks)	Incomplete justification of the objectives	Only some objectives of the Proposed work

				proposed. (2 Marks)	are well defined. (1 Mark)
3	Literature survey (5)	Relevant Information is gathered from multiple, research-based sources. Detailed conclusions are reached from the evidence offered (5 Marks)	Relevant Information is gathered from a limited number of sources. There is some indication of conclusions from the evidence offered. (3 Mark)	--	--
4.	Methodology of Project Proposal (8)	Steps to be followed to solve the defined problem are clearly specified. (8 Marks)	The methodology to be followed is specified but no detailed explanation. (6 Marks)	Steps are mentioned but unclear, without justification for objectives (3 Marks)	The steps to be followed to solve the defined problem are not specified properly. (2 Mark)
5.	Communication Skills (5)	The entire audience can hear the presentation; maintains eye contact with the audience; has a clear, expressive voice; is poised, has good posture, and has no distracting mannerisms. (5 Marks)	Most of the audience can hear the presentation; eye contact most of the time; clear voice, but not as expressive; a little nervous, not as polished. (3 Marks)	--	Difficult to hear; occasional eye contact; some mumbling, little or no expression; nervous, some distracting mannerisms; reads much of slide. (1 Mark)
6.	Quality, appropriateness, and accuracy of project Implementation (8)	Has decided on the relevant tools and platforms required for the project by evaluating the alternatives. Has coded all the components designed by the student by following the standard coding guidelines. The code is completely documented with comments. (8 marks)	Has coded all the components designed by the student by following the standard coding guidelines. The code is completely documented with comments. (6 marks)	Has coded more than 80% of the components designed by the student but has not followed the standard coding guidelines. The code is not documented well with comments. (4 marks)	Has coded less than 50% of the components designed by the student but has not followed the standard coding guidelines. The code is not documented with comments. (2 marks)
7.	Interpreting the findings(5 marks) Or Deliverables	Student provides a logical interpretation of the results and findings and clearly solves the problem (5 marks)	The student provides a logical but incomplete interpretation of the results and findings and clearly solves the problem (3 marks)	The student does not provide an interpretation of the results and findings but solves the problem. (2 marks)	The student does not interpret the findings / reach a conclusion (1 mark)

8.	Individual Contribution (5 marks)	Excellent Contribution showing his/her dependency on the project (5 Marks)		Some contributions as reflected in the overall work. (2 Mark)	
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Performance Indicators for SEE	Marks(50 M)
Write – Up	10
Demonstration + Presentation	30
Viva-Voce	10

TECHNICAL SKILL ENHANCEMENT COURSE	
Course Code: IS68	Credits: 0:1:0
Pre – requisites: Operating system, DBMS, Computer Networks, Software Design Patterns, Object Oriented Concepts and Data Structures	Contact Hours: 28 hours
Course Coordinator: Dr. Lincy M M	

Course Contents

Week 1 & 2: Data Structures & Algorithms

Quiz, Assignments, Project Based Learning on:

- Complexity, Arrays & Strings (sliding window, hashing, two pointers), Trees & Graphs (Algorithms), Dynamic Programming (Algorithms), Segment Trees, Fenwick Trees, Tries, Greedy vs Divide & Conquer vs DP Comparison.

Recommended Resources / Links/ Assignments:

- LeetCode (Medium/Hard)
- GeeksforGeeks Practice
- Codeforces contests

Week 3 & 4: Computer Networks

Quiz, Assignments, Project Based Learning on:

- OSI vs TCP/IP, TCP vs UDP (congestion control, flow control, 3-way handshake, Routing protocols and algorithms, Subnetting, IPv4 vs IPv6., HTTP/1.1 → HTTP/2 → HTTP/3, Network Security basics (SSL/TLS, firewalls).

Recommended Resources / Links:

- Wireshark packet analysis
- Cisco Packet Tracer labs
- Linux commands: netstat, ping, ss, traceroute

Week 5 & 6 : Database Management Systems

Quiz, Assignments, Project Based Learning on:

- Advanced SQL , Transactions & ACID, Indexing & Query optimization, Normalization vs Denormalization, Concurrency control: 2PL, Deadlocks, Recovery, Triggers, Stored Procedures, NoSQL basics

Recommended Resources / Links:

- LeetCode DB problems
- HackerRank SQL challenges
- DB-Fiddle experiments

Week 7: Object-Oriented Programming

Quiz, Assignments, Project Based Learning on:

- Abstraction, Encapsulation, Inheritance, Polymorphism, Method Overloading vs Overriding, Abstract class vs Interface, Multiple inheritance, Static vs Dynamic binding, Heap vs Stack memory in OOP

Recommended Resources / Links:

- GFG OOP Interview Questions
- Implement mini-OOP design case studies in Java/Python

Week 8: Software Design Patterns

Quiz, Assignments, Project Based Learning on:

- Creational: Singleton, Factory, Structural: Adapter, Decorator, Behavioral: Observer, Strategy, SOLID principles, Clean code best practices, Low-level design

Recommended Resources / Links:

1. Refactoring Guru implementations
2. GitHub System Design Primer

Week 9–10: Operating Systems (Linux + Kernel Focus)

Quiz, Assignments, Project Based Learning on:

- Kernel: Monolithic vs Microkernel, System calls, Device drivers, Kernel space vs User space, Process Management: Scheduling, Memory Management: Paging, Segmentation, Virtual memory, Thrashing, Deadlocks, File Systems
- Linux/Ubuntu Specific:
 - Commands: ps, top, kill, chmod, chown, df, du, grep, awk, sed, nohup, jobs, fg/bg.
 - Signals: SIGKILL, SIGTERM, SIGSTOP.
 - Hard vs Soft links.
 - Package management: apt-get, dpkg.
 - Logs: dmesg, /var/log/.

Recommended Resources / Links:

1. Implement fork/exec programs in C.
2. Bash scripting basics.
3. Debugging with gdb on Linux.

Weeks 11–14: Capstone Projects

Quiz, Assignments, Project Based Learning on:

- Students to work on projects covering the topics mentioned above.

TUTORIALS

Assigned by the handling faculty each week.

Reference Books:

1. Aaron M. Tanenbaum, Yedidyah Langsam and Moshe J. Augenstein, “Data Structures Using C”, 2nd Edition, PHI
2. Reema Thareja, S. Rama Sree, Advanced Data Structures, Oxford University Press, 2017
3. Tanenbaum Computer Networks, Pearson Education; Sixth edition, 2022.
4. E Balagurusamy Object Oriented Programming With C++, 8th Edition
5. Erich Gamma. Design Patterns: Elements of Reusable Object Oriented Software, 2015
6. Rajiv Chopra. Operating Systems: A Practical Approach. S Chand Publishing; Fourth edition

Course Outcomes:

1. Apply core concepts of Data Structures, Computer Networks, Database Management Systems, Operating Systems, Object-Oriented Programming, and Software Design Patterns for real world computing problems. (PO -1,2,3,4,5,6,7,8,9,10,11) (PSO -1,2,3)
2. Exhibit problem-solving, critical thinking, and technical communication skills by engaging in coding challenges, assignments, and project-based learning activities aligned with professional practices (PO -1,2,3,4,5,6,7,8,9,10,11) (PSO -1,2,3)
3. Implement theoretical knowledge to practical and industry-relevant challenges. (PO -1,2,3,4,5,6,7,8,9,10,11) (PSO -1,2,3)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Quiz, Assignments, Project Based Learning Assessment	30	CO1, CO2
Other Components		
Project Evaluation	20	CO3